

Sir2 and Fun30 regulate ribosomal DNA replication timing via Mcm helicase positioning and nucleosome occupancy

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Abstract

The association between late replication timing and low transcription rates in eukaryotic heterochromatin is well-known, yet the specific mechanisms underlying this link remain uncertain. In *Saccharomyces cerevisiae*, the histone deacetylase Sir2 is required for both transcriptional silencing and late replication at the repetitive ribosomal DNA arrays (rDNA). We have previously reported that in the absence of *SIR2*, a derepressed RNA PolII repositions MCM replicative helicases from their loading site at the ribosomal origin, where they abut well-positioned, high-occupancy nucleosomes, to an adjacent region with lower nucleosome occupancy. By developing a method that can distinguish activation of closely spaced MCM complexes, here we show that the displaced MCMs at rDNA origins have increased firing propensity compared to the non-displaced MCMs. Furthermore, we found that both, activation of the repositioned MCMs and low occupancy of the adjacent nucleosomes critically depend on the chromatin remodeling activity of *FUN30*. Our study elucidates the mechanism by which Sir2 delays replication timing, and it demonstrates, for the first time, that activation of a specific replication origin *in vivo* relies on the nucleosome context shaped by a single chromatin remodeler.

eLife assessment

This study is a detailed investigation of how chromatin structure influences replication origin function in yeast ribosomal DNA, with focus on the role of the histone deacetylase Sir2 and the chromatin remodeler Fun30. **Convincing** evidence shows that Sir2 does not affect origin licensing but rather affects local transcription and nucleosome positioning which correlates with increased origin firing. However, the evidence remains **incomplete** as the methods employed do not rigorously establish a key aspect of the mechanism, fully address some alternative models, or sufficiently relate to prior results. Overall, this is a **valuable** advance for the field that could be improved to establish a more robust paradigm.

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Introduction

SIR2 is the founding member of a family of NAD-dependent protein deacetylases called sirtuins, which have important effects on transcription, replication, chromatin structure and metabolism in organisms from yeast to humans (Houtkooper et al. 2012 [↗](#); Shahgaldi and Kahmini 2021 [↗](#)). *SIR2* was initially identified in budding yeast in 1979 on the basis of its role in transcriptional repression at the silent mating type loci (Haber and George 1979 [↗](#); Klar et al. 1979 [↗](#); Rine et al. 1979 [↗](#)), and it was therefore dubbed a “Silent Information Regulator”. The silent mating type loci are heterochromatic regions that contain non-transcribed copies of the small open reading frames that determine yeast mating type, and deletion of *SIR2* not only activates their transcription but also abolishes their heterochromatic structure (Ivy et al. 1986 [↗](#); Rine and Herskowitz 1987 [↗](#); Weiss and Simpson 1998 [↗](#); Ravindra et al. 1999 [↗](#); Haber 2012 [↗](#)). It was later discovered that, in addition to regulating transcription at the silent mating type loci, Sir2 also regulates DNA replication, and it does so not only at the silent mating type loci but also at the two other heterochromatic regions of the yeast genome, namely at telomeres and the rDNA (Pasero et al. 2002 [↗](#); Yoshida et al. 2014 [↗](#)). It is this last activity of Sir2, namely its effect on replication at the rDNA, that is the focus of this report.

The rDNA, which encodes the structural RNA components of the ribosome, is composed of approximately 150 tandemly arranged 9.1 kb repeats (Woolford and Baserga 2013 [↗](#)). At almost 1.4 megabases, the rDNA is too large to be replicated passively, and thus it must contain its own origins of replication. This is achieved by the organization shown in **Figure 1A** [↗](#): The ribosomal RNAs are produced as two major species, known as the 35S and 5S, which are transcribed by RNA polII and RNA polIII, respectively (Fernandez-Pevida et al. 2015 [↗](#)). Between the 5' ends of these genes is an origin of replication, and immediately to the left of this origin is a short Sir2-repressed RNA PolII-transcribed non-coding transcript called C-pro (Kobayashi and Ganley 2005 [↗](#); Li et al. 2006 [↗](#); Vasiljeva et al. 2008 [↗](#)), which we will discuss further below. In wild-type cells, rDNA origins fire in the second half of S phase, making the rDNA among the later replicating regions of the genome, while in *sir2* mutants, they fire early in S phase (Pasero et al. 2002 [↗](#); Yoshida et al. 2014 [↗](#); Foss et al. 2017 [↗](#)).

An indication of the biological significance of maintaining the late replication status of the rDNA came from a genome wide association study aimed at identifying naturally occurring genetic variants that affect replicative lifespan (Kwan et al. 2013 [↗](#)). Replicative lifespan in budding yeast is defined as the number of daughter cells that a mother cell is able to produce and is therefore a measure of a cell's replicative capacity at birth. By measuring lifespan in segregants from a cross between two genetically diverse strains of yeast, Kwan et al. were able to identify a point mutation at the *rARS* that (1) decreased origin activity at the rDNA; (2) increased origin activity elsewhere in the genome; and (3) prolonged replicative lifespan. They interpreted these results as reflecting competition for replication resources between the origins in two distinct compartments of the nucleus, namely the heterochromatic repetitive nucleolus or rDNA and the non-repetitive sequences that make up the bulk of the yeast genome. This competition is exacerbated by the perturbation of nucleolar heterochromatin caused by the absence of *SIR2* (Yoshida et al. 2014 [↗](#); Foss et al. 2017 [↗](#)). Most replication outside of the rDNA initiates from approximately 300 replication origins, and thus the 150 origins at the rDNA make up approximately one third of total origins. Increased replication resources directed toward the rDNA might therefore be expected to create a significant strain on replication in the rest of the genome.

Replication origins are known to compete for limiting resources at the level of origin firing. Broadly speaking, replication origin activity is regulated at two levels, namely origin licensing and origin firing (Remus and Diffley 2009 [↗](#)). Origin licensing refers to loading of the replicative helicase in G1, and only origins that have been licensed are capable of being used in the

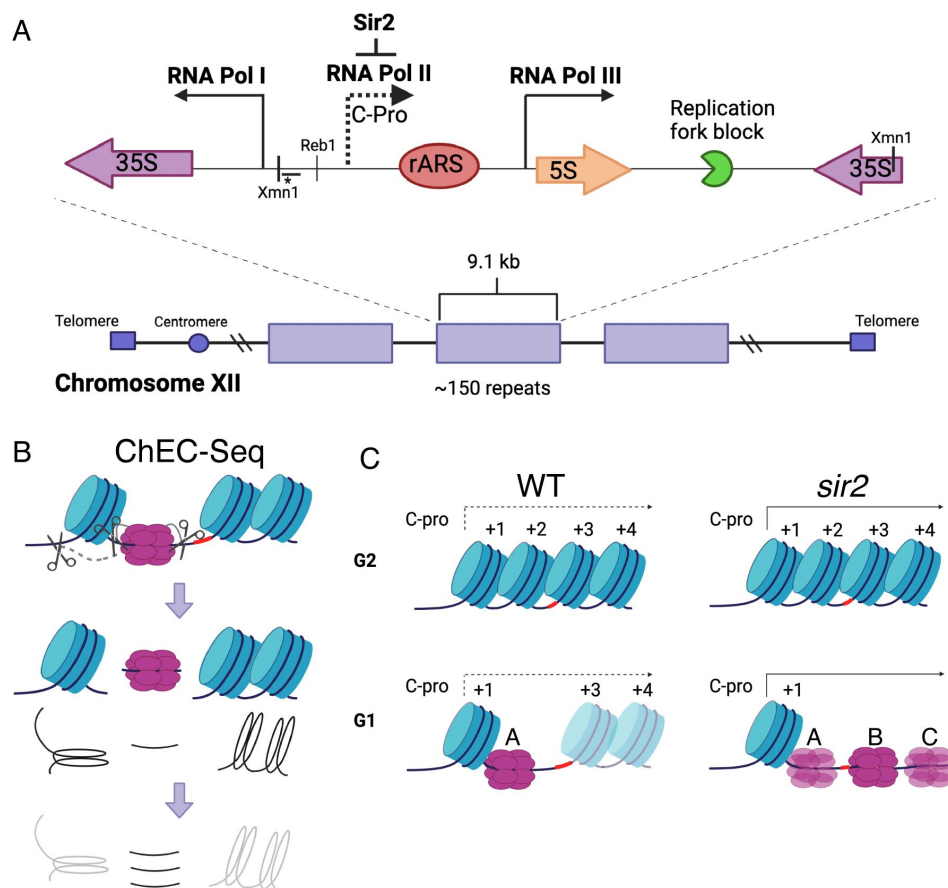


Figure 1.

rDNA structure, Chromatin Endogenous Cleavage (ChEC), and Mcm displacement in *sir2*.

A. The 1.4 megabase rDNA region on chromosome XII is composed of approximately one hundred and fifty 9.1 kb repeats. Each repeat encodes both the 35S and 5S ribosomal RNAs (rRNA), which are transcribed by RNA Polymerase I (Pol I) and RNA Polymerase III (Pol III), respectively. The ribosomal origin of replication (rARS) is located between the 5' ends of these genes. The C-pro transcript, which is suppressed by Sir2, initiates approximately two hundred base pairs from the rARS. The unidirectional replication fork block, which functions to prevent collision between replication and transcription machinery, is depicted in green. The probe used in Southern blots in [Figure 3](#) to assess licensing is marked by *. **B.** In ChEC-seq, micrococcal nuclease (MNase; depicted as scissors) is fused to the protein of interest, in this case Mcm2. The double-hexameric Mcm helicase complex, Mcm2-7, is depicted in purple, and nucleosomes are in blue. Cleavage is induced by addition of calcium to permeabilized cells. Due to the proximity of nucleosome entrance and exit sites, Mcm-ChEC can reveal not only the binding site of the Mcm complex but also that of the adjacent nucleosome. **C.** Depiction of nucleosomes and Mcm complex in wild type and *sir2* in G1 and G2. Nucleosomes are numbered with respect to C-pro transcription. In WT cells, Mcm is found almost exclusively at a single position, labeled as "A". Derepression of C-pro transcription in *sir2* causes RNA PolII to displace the Mcm complex rightward to positions "B" and "C". The three different Mcm helicase complexes depicted in the bottom right panel are intended to convey the three most prominent locations for the complex in *sir2*; this is not intended to indicate that there is ever more than one Mcm complex in the same rDNA repeat.

subsequent S phase. Licensing occurs when an Origin Recognition Complex, Orc1-6, that is bound to an 11-17 base pair so-called ARS consensus sequence (ACS) within the origin loads two hexameric helicases, each consisting of a ring of subunits, Mcm2-7, that encircles the DNA (Bell and Stillman 1992 [↗](#); Francis et al. 2009 [↗](#); Bell and Botchan 2013 [↗](#); Deegan and Diffley 2016 [↗](#); Deegan et al. 2016 [↗](#); Douglas et al. 2018 [↗](#); Lewis et al. 2022 [↗](#)). The mechanism by which Orc recognizes origins and loads an Mcm double-hexamer has been elucidated in great detail (Bell and Labib 2016 [↗](#); Costa and Diffley 2022 [↗](#); Hu and Stillman 2023 [↗](#)). DNA replication initiates from only a subset of these licensed origins, namely those where the helicase complex is activated, a process referred to as origin firing (Rhind and Gilbert 2013 [↗](#)). Origin firing is triggered by activation of two kinase signaling pathways, DDK and CDK, which then cause each hexameric Mcm helicase to encircle a single strand of DNA and track in opposite directions in front of the replisome, thereby establishing bidirectional replication (Heller et al. 2011 [↗](#); Larasati and Duncker 2016 [↗](#); Li et al. 2023 [↗](#)). Firing factors are not present in sufficient quantities to activate all licensed origins simultaneously, and thus origins are activated in waves, with firing factors being recycled until the entire genome has been replicated (Mantiero et al. 2011 [↗](#); Tanaka et al. 2011 [↗](#)). Determining the criteria by which origins are prioritized for activation remains an area of active research.

Importantly, if the competition between the origins at the rDNA and those in the rest of the genome is tilted in favor of the rDNA, either by a point mutation in the rARS or by deletion of *SIR2*, replication gaps between distantly spaced origins outside of the rDNA persist late into S phase, and genome duplication may remain incomplete when the cell enters mitosis (Foss et al. 2017 [↗](#)). Indeed, the observation that weakening the rARS by a point mutation can suppress the short replicative lifespan characteristic of *sir2* mutants suggests that these replication gaps may be the ultimate cause of “replicative death”. Teleologically, one can rationalize the prioritization of replication of the unique areas of the genome over the repetitive and heterochromatic areas, because (1) there are mechanisms available for repair in repetitive regions, like single-strand annealing, that are not available in unique regions, and they may therefore be able to survive lesions created when mitosis precedes the completion of replication; and (2) the cell is able to tolerate variation in copy number at the rDNA, but not deletion of protein-coding sequences in unique regions. Consistent with this view of the utility of assigning a low priority to replication of repetitive sequence, heterochromatic regions are also late-replicating in mammalian cells, and disease states have been associated with disruption of this pattern (Donley and Thayer 2013 [↗](#)).

Despite its biological importance, the mechanism by which Sir2 suppresses replication initiation is poorly understood. Clues to Sir2’s role in replication may be gleaned from its role in transcription, although here, too, our understanding is limited. Sir2’s ability to repress transcription at the silent mating type loci has been postulated to follow from its role in chromatin compaction. Experiments demonstrating *SIR2*-dependent protection of heterochromatin from digestion with endonucleases or methylation by bacterial DNA methyltransferases support the notion that DNA in heterochromatin is generally less accessible (Gottschling 1992 [↗](#); Singh and Klar 1992 [↗](#); Loo and Rine 1994 [↗](#); Ansari and Gartenberg 1999 [↗](#)). Thus, a widely publicized model attributes Sir2’s ability to inhibit transcription to blocking access of RNA PolII to the DNA, although a simple model of indiscriminate steric occlusion cannot account for this phenomenon (Chen and Widom 2005 [↗](#); Lynch and Rusche 2009 [↗](#); Kitada et al. 2012 [↗](#); Bondra and Rine 2023 [↗](#)). Whether Sir2’s effect on replication is similarly the result of chromatin compaction limiting access to enzymes involved in DNA metabolism is not known.

A study examining the distribution of Mcm helicase complexes at the rDNA suggests that, rather than a global manifestation of lack of accessibility of the rDNA, the early activation of replication at the rDNA in the absence of Sir2 may reflect a more localized phenomenon (Foss et al. 2019 [↗](#)). In this study, Mcm binding was assessed at nucleotide-level resolution by fusing a subunit of Mcm to micrococcal nuclease (MNase), a technique known as Chromatin Endogenous Cleavage, or “ChEC” (Figure 1B [↗](#)) (Schmid et al. 2004 [↗](#); Zentner et al. 2015 [↗](#)). Permeabilization of cells with

detergent followed by addition of calcium activates MNase, thereby generating approximately 65 base-pair fragments of DNA under Mcm double-hexamers that can be identified by high-throughput sequencing. This analysis revealed that, while deletion of *SIR2* did not elicit an obvious change in the overall levels of licensing at the rDNA, it led to a dramatic redistribution of Mcm helicase complexes: In wild-type cells, the helicase complexes are located almost exclusively immediately adjacent to a high occupancy well-positioned nucleosome (+1 nucleosome in **Figure 1C**) approximately 130 base pairs to the left of the ACS (position A in **Figure 1C**). In contrast, the Mcm complexes in *sir2* mutants are largely displaced to the right into an area with lower nucleosome occupancy (positions B and C in **Figure 1C** and **Figure 1-figure supplement 1**). This displacement appears to be caused by the de-repression of C-pro transcription in the absence of *SIR2*, which results in RNA PolII pushing the Mcm double-hexamer into the relatively nucleosome-free region. This observation is provocative in light of studies demonstrating that the arrangement of Mcm-proximal nucleosomes can have a dramatic effect on Mcm activation (Azmi et al. 2017). Specifically, by using chromatin remodeling enzymes (CREs) to create a variety of nucleosomal arrangements on an origin-containing fragment of DNA, Azmi et al. demonstrated that activation of an Mcm double-hexamer loaded at that origin could be inhibited by nucleosomes arranged by specific CREs. In another study (Chacin et al. 2023), a mutant version of Orc1 that retains the ability to load Mcm but alters phasing of Mcm adjacent nucleosomes was shown to inhibit DNA replication *in vivo* and *in vitro*, consistent with a critical role for nucleosome organization at origins in the initiation of replication.

Taken together, the observations above led us to hypothesize that (1) relocation of Mcm helicases from regions of high nucleosome occupancy to nearby regions of low nucleosome occupancy promotes early activation of replication origins in *sir2*; (2) this local difference in nucleosome occupancy is the result of modulation of nucleosomal architecture by one or more CREs; and therefore (3) screening CREs for the ability to suppress the early rDNA replication phenotype in *sir2* mutants might allow us to identify the relevant CRE. In this report, we show that the displaced Mcm complexes fire before their non-displaced counterpart and that increased nucleosome occupancy near the displaced Mcms in the absence of Fun30 CRE suppresses their early activation. This is the first *in vivo* demonstration of the modulation of an origin activity by a specific chromatin remodeler *in vivo*, and has implications for the broader relationship between transcription and replication timing.

Results

***Fun30* chromatin remodeling activity is required for early rDNA replication and for maintaining the rDNA array size in *sir2* mutant**

To identify CREs required for the early replication of the rDNA in *sir2* mutants, we needed a method to determine replication timing, and for this we turned to S-seq. In this widely used technique, G1 and S phase populations of cells from a log phase culture are sorted using flow cytometry and their DNA subjected to genomic sequencing (Muller et al. 2014; Batrakou et al. 2020). Replication timing can then be inferred from read depths in the S phase fraction, because regions that replicate early in S phase will be present in two copies in most cells, while regions that replicate late will be present mostly as a single copy. S phase read depths are normalized to those in G1 to account for differences in DNA copy numbers, which is particularly important for repetitive loci whose copy number can vary, such as rDNA. As shown in **Figure 2A**, which depicts the S-seq profile of chromosome 12, relative replication timing (Trel) values using this approach typically range from 1.4 to 0.8 for early- and late-replicating regions, respectively, with a value of 1 indicating that a locus replicates at the same time as the genome-wide average (see Materials and Methods for specific calculation). The late replication of the rDNA, defined as coordinates 450,000 - 470,000 on chrXII, in wild type is reflected in a Trel value of 0.89, whereas this ratio increases to 1.2 upon deletion of *SIR2* (**Figure 2A,B**). Using this method, we showed

that a point mutation in the ACS of the rARS suppresses early replication of the rDNA in *sir2*, while promoting replication in the rest of the genome (Kwan et al. 2013 [↗](#); Foss et al. 2017 [↗](#)). To identify potential chromatin remodelers required for early rDNA replication in *sir2*, we measured rDNA replication timing in double-mutants in *sir2* and each of 10 non-essential CRE subunits (Chd1, Fun30, Htl1, Irc5, Isw1, Isw2, Nhp10, Swp82). Deletion of *FUN30*, but no other gene, delayed replication of the rDNA in a *sir2* background, shifting the Trel value from 1.2 to 1.06, while having had no significant effect in *SIR2* cells (0.89 for WT and 0.91 for *fun30*) (**Figure 2B** [↗](#)).

Because Fun30 can form a complex with the replication scaffold protein Dpb11, it was possible that Fun30 played a structural rather than an enzymatic role in regulating rDNA replication. However, we found that a point mutation that abolishes Fun30 ATPase activity, K603R (Neves-Costa et al. 2009 [↗](#); Eapen et al. 2012 [↗](#)), also suppresses early rDNA replication (**Figure 2B** [↗](#)), arguing that Fun30 instead promotes rDNA replication through its enzymatic activity in nucleosome remodeling. Fun30 is also known to play a role in the DNA damage response; specifically, phosphorylation of Fun30 on S20 and S28 by CDK1 targets Fun30 to sites of DNA damage, where it promotes DNA resection (Chen et al. 2016 [↗](#); Bantele et al. 2017 [↗](#)). To determine whether the replication phenotype that we observed might be a consequence of Fun30's role in the DNA damage response, we tested non-phosphorylatable mutants for the ability to suppress early replication of the rDNA in *sir2*; these mutations had no effect on the replication phenotype (**Figure 2B** [↗](#)), arguing against a primary role for Fun30 in DNA damage repair that somehow manifests itself in replication.

We next wanted to determine whether *FUN30* affects rDNA array size. The repetitive nature of the rDNA allows it to fluctuate in size, and mutations that impair DNA replication, either globally or in a manner specific to the rDNA, are associated with shrinkage of the rDNA. For example, the rDNA in a wild-type strain in the S288c background consists of 150 copies, but a point mutation at the ARS consensus sequence (ACS) that impairs Orc binding causes the rDNA to shrink to just 80 copies (Kwan et al. 2013 [↗](#)). rDNA shrinkage in response to global replication defects is thought to reflect the fact that the rDNA, which typically comprises approximately 10% of genomic DNA, imposes a significant replicative burden on the cell that can be ameliorated by decreasing the number of repeats (Salim et al. 2017 [↗](#)).

We determined rDNA size by assessing the fraction of DNA sequencing reads from a sonicated G1 population that arose from the rDNA. While deletion of *FUN30* had no notable effect on rDNA size in a *SIR2* strain, it caused a 50% reduction in a *sir2* mutant, from 9.2% to 4.2% (**Figure 2C** [↗](#)). rDNA size reduction of a similar magnitude was observed in the ATPase-deficient Fun30K603R mutant, but not in the non-phosphorylatable Fun30S20AS28A mutant, mirroring the disparate effect of the two alleles on rDNA replication timing. The reduction of the rDNA size in *sir2 fun30* mutants was confirmed by qPCR analysis of DNA isolated from log cultures, and by Southern blots (**Figure 2-figure supplement 1** [↗](#) and **Figure 3** [↗](#)). Based on measurements in log-growing cells, we estimated the rDNA size in *fun30 sir2* mutants to ~40 repeats (**Figure 2-figure supplement 1** [↗](#)). This places it in a category comparable to the shortest rDNA arrays in a collection of DNA replication mutants noted for their contraction of the rDNA (Salim et al. 2017 [↗](#)), thus emphasizing the severity of the replicative disadvantage conferred by the deletion of *FUN30* when Sir2 function is absent.

In the analysis above, we are assuming that the reduced rDNA size is a downstream effect of decreased rDNA origin activity. While there is ample support for this view of cause and effect (Kwan et al. 2013 [↗](#); Salim et al. 2017 [↗](#)), rDNA shrinkage itself has been reported to increase rDNA origin activity (Kwan et al. 2023 [↗](#)). Such an effect would be expected to counteract the rDNA replication suppression conferred by *FUN30* deletion, and it therefore suggests an explanation for our observation that rDNA replication timing in *sir2 fun30* cells is not suppressed to the level observed in WT cells.

Figure 2.

S-Seq applied to rDNA replication timing and copy number determination.

A. S-seq replication profile of Chromosome XII. The region in the middle noted as the rDNA has been collapsed. Note that the rDNA replicates much earlier in *sir2* (orange) than in wild type (blue). **B.** rDNA replication timing in double mutants between *sir2* and various chromatin remodeling enzymes. *fun30-S20A S28A* is non-phosphorylatable, and *fun30-K603R* is catalytically inactive. The S to G1 ratio values (mean $\pm$ SD) were 0.90 $\pm$ 0.02 for WT, 1.04 $\pm$ 0.04 for *fun30sir2*, 1.01 $\pm$ 0.02 for *fun30K603R* and 1.21 $\pm$ 0.03 for *sir2* ($p < 0.001$ for WT vs *sir2*, *sir2* vs *fun30sir2* and *sir2* vs *fun30K603R* *sir2* using t-test). **C.** Effect of *sir2* and *fun30* mutation on rDNA size, as determined from the fraction of G1 sequencing reads that arise from 450-470 kb on chrXII. The values (% mean $\pm$ SD) were 9.4 $\pm$ 1.6 for WT, 4.6 $\pm$ 1.2 for *fun30sir2*, 4.8 $\pm$ 0.3 for *fun30K603R* and 9.2 $\pm$ 1.8 for *sir2* ($p < 0.001$ for WT vs *sir2*, *sir2* vs *fun30sir2* and *sir2* vs *fun30K603R* *sir2* using t-test)

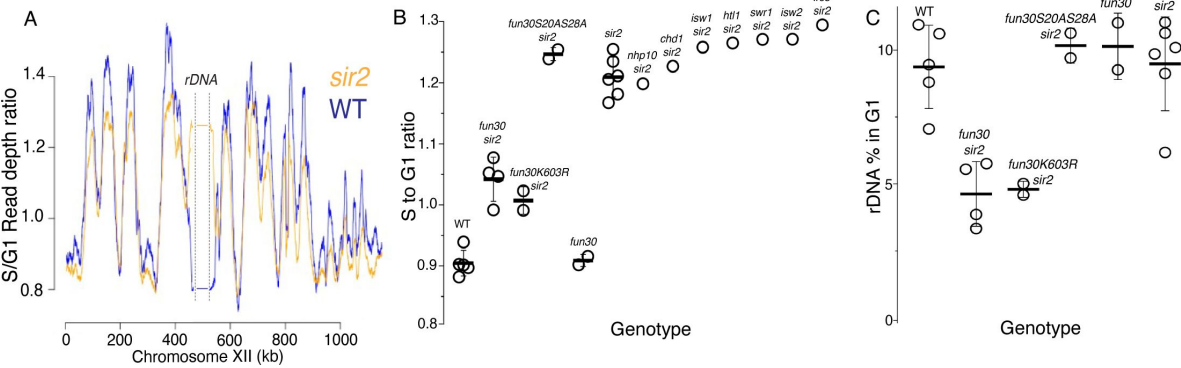
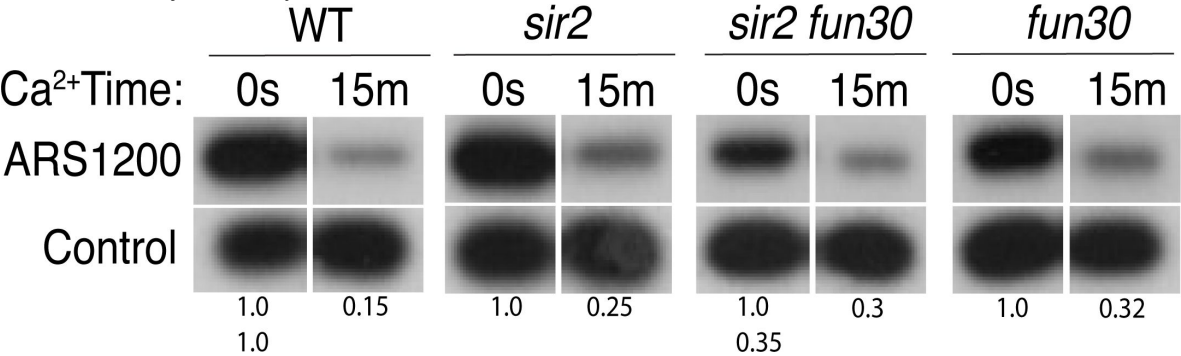


Figure 3.

Licensing at the rARS, as determined by Southern blot.

Activation of Mcm-MNase at the rARS with calcium will eliminate the XmnI fragment, as denoted in **Figure 1**. *PIK1*, a single-copy gene in which we detect no Mcm binding, is used as a control. Quantification of the uncut band was used to infer relative rDNA array size in *sir2 fun30* mutant.



In summary, we conclude that Fun30 nucleosome remodeling activity promotes rDNA replication and is required for maintaining rDNA size in *sir2* mutants. We next aimed to determine whether Fun30 exerts this effect by reducing Mcm loading or the ability of loaded MCMs to fire.

High level of origin licensing at rDNA origins in WT, *sir2* and *sir2 fun30* mutants

It has been estimated that ~20% of potential rDNA origins are activated during any single S-phase in wild-type cells (Brewer and Fangman 1988 [↗](#); Linskens and Huberman 1988 [↗](#)). Furthermore, single-molecule studies suggested that the fraction of active origins is increased in *sir2* mutant cells (Pasero et al. 2002 [↗](#); Yoshida et al. 2014 [↗](#)). To determine whether *SIR2* and *FUN30* modulate rDNA replication by changing origin licensing, i.e. the fraction of origins that are bound by Mcm helicase complexes prior to the beginning of S phase, we assessed licensing in WT, *sir2*, *fun30*, and *sir2 fun30* cells by measuring DNA cleavage by Mcm2-MNase using Southern blots (Foss et al. 2021 [↗](#)): Digestion with XmnI yields a 3.5 kb fragment that spans the rARS (Figure 1A [↗](#)), and, if the rARS is licensed, activation of MNase by addition of calcium will eliminate this fragment using a probe upstream of the rARS. As a control, we used a fragment that we know does not contain Mcm complexes. Activation of MNase in all 4 genotypes led to an ~70-85% decrease in the XmnI fragment (Figure 3 [↗](#)), therefore we conclude that (1) licensing is high, at ~70-85%; and (2) a much higher fraction of origins are licensed than are activated, making it unlikely that regulation of origin activity occurs at the level of licensing. This is the first quantitative assessment of licensing at the rDNA in either wild type or *sir2*.

Fun30 promotes firing of rDNA origins and suppresses firing of non-rDNA origins in *sir2* mutants

Our observation that *FUN30* status had no obvious effect on rDNA origin licensing suggested that Fun30 might instead advance rDNA replication in *sir2* mutants by promoting rDNA origin firing. To evaluate this possibility we measured the effect of *FUN30* status on origin activity at the rDNA using 2D gels. In this technique, active origins generate replication bubble signals, whereas passive replication of an origin appears as a y arc, and origin activity can therefore be assessed as the ratio of bubble-to-y arcs. We grew cells to log phase, arrested them for 1.5 hours in alpha factor, and then released them into 200 mM hydroxyurea (HU). As seen previously, deletion of *SIR2* advances replication at the rDNA (for example, compare the bubble-to-y-arc ratio in WT and *sir2* at 30 minutes (Figure 4A [↗](#)), while deletion of *FUN30* suppresses this effect. Deletion of *FUN30* in a strain with a wild-type copy of *SIR2* had no obvious effect. We conclude that deletion of *FUN30* partially reverses the accelerated activation of the ribosomal origins caused by deletion of *SIR2*.

We and others have previously shown that deletion of *SIR2* not only advances replication at the rDNA but also retards replication elsewhere in the genome (Kwan et al. 2013 [↗](#); Yoshida et al. 2014 [↗](#)). The absence of *SIR2* has been linked to decreased activation of less robust early replication origins, likely due to their heightened sensitivity to overall levels of replication-initiating factors. To ascertain the specificity of the interaction between *FUN30* and *SIR2*, we investigated whether deleting *FUN30* also mitigates the delayed activation of weak early origins in a *sir2* mutant background. We focused on the activation of ARS305, an origin we previously identified as delayed in a *sir2* mutant using 2D gel analysis (Foss et al. 2019 [↗](#)). As seen previously, deletion of *SIR2* suppresses ARS305, while additional deletion of *FUN30* restores the bubble-to-arc ratio to near that of wild type (Figure 4B [↗](#)). Deletion of *FUN30* in a *SIR2* background had no perceptible effect in this assay.

Delayed initiation of genomic origins following *SIR2* deletion has been previously demonstrated on a global scale as a decline in the incorporation of a labeled uridine analogue, bromodeoxyuridine, at weak early origins in *sir2* cells compared to wild-type cells upon their release from G1 into HU (Yoshida et al. 2014 [↗](#)). We employed a similar approach to investigate

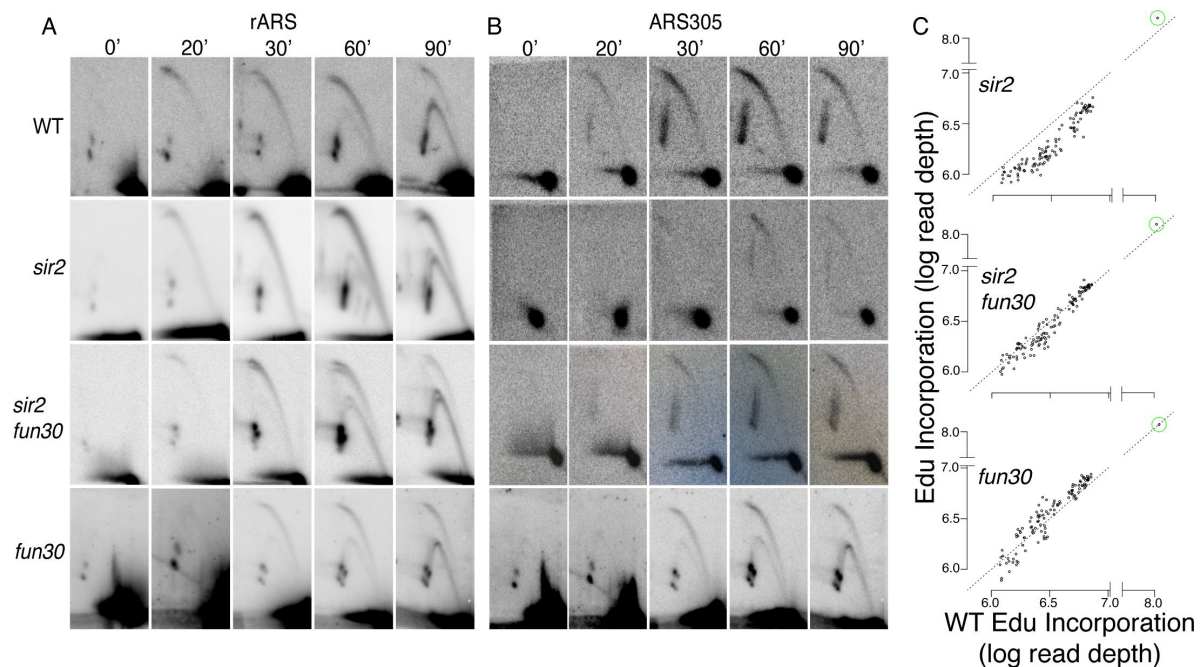


Figure 4.

Deletion of FUN30 suppresses origin activity at rDNA and promotes it elsewhere in the genome.

A. Cells were arrested in G1 with alpha factor and then released into medium containing 200 mM HU. The ratio of bubble to y arc in 2D gels demonstrates increased rARS activity in *sir2* that is suppressed by *fun30*. **B.** Activation of the rARS in *sir2* is accompanied by suppression of ARS305, and this effect is likewise reversed by additional deletion of *fun30*. **C.** EdU incorporation at 111 early origins demonstrates suppression of origin activity by deletion of *sir2* and suppression of this effect by *fun30*. Each dot represents a single origin, quantified according to EdU incorporation 1 hour after release from G1 into medium containing 200 mM HU. EdU incorporation was summed across a 5 kb window centered on the Mcm binding site within each origin. Points are plotted according to EdU signal in wild type on the x axis and in the mutant in question on the y axis. Suppression of origin activity is reflected in points dropping below the dotted line at 45°. The rARS is circled in green.

whether the deletion of *FUN30* mitigates this effect, utilizing a different labeled nucleotide, ethynyl-deoxy-uridine (EdU), which can be isolated through click chemistry rather than immunoprecipitation. Wild-type (WT), *sir2*, *sir2 fun30*, and *fun30* cells engineered for the uptake and phosphorylation of the analogue (Viggiani and Aparicio 2006 [↗](#)) were arrested in G1 using alpha factor and collected 60 minutes after release into medium containing HU and EdU. DNA extracted from these cells was sonicated and subjected to click chemistry with biotin azide to covalently link biotin to EdU incorporated into DNA. Biotinylated DNA was then purified and sequenced. The impact of *SIR2* and *FUN30* status on replication timing was assessed by graphing the EdU signal at 111 early origins in WT cells on the x-axis against the signal in *sir2*, *fun30*, and *sir2 fun30* cells on the y-axis (**Figure 4C** [↗](#)). Similar to observations with BrdU, we noted reduced EdU incorporation at a subset of early origins in *sir2* mutants, coupled with increased EdU incorporation at rDNA origins. Deletion of *FUN30* in a *sir2* background partially restored EdU incorporation at early origins, concomitant with reduced EdU incorporation at rDNA origins. However, *FUN30* deletion in the WT background did not significantly impact EdU incorporation at either rDNA or genomic origins. These findings highlight that *FUN30* deletion-induced suppression of rDNA origins is accompanied by the activation of genomic origins.

Early firing of displaced helicase complexes

While our results above demonstrate that deletion of *SIR2* promotes firing of the rDNA origins and that additional deletion of *FUN30* suppresses this phenotype, they lack the resolution to distinguish replication arising from the displaced versus the non-displaced Mcm complexes. Other methods for measuring origin activation, such as generation of single-stranded DNA or incorporation of nucleoside analogs, also suffer from this limitation. Distinguishing between firing of the displaced versus the non-displaced Mcm complexes is important to test our hypothesis that displacement of the helicase complex into a relatively nucleosome-free region is responsible for its early activation. We therefore sought to devise a method that can differentiate activation of Mcm complexes that are only ~150 bp apart, i.e. the approximate distance between displaced and non-displaced Mcms. We reasoned that, because MCM-ChEC has the resolution to distinguish closely spaced helicase complexes, the differential disappearance of such closely spaced footprints might permit us to distinguish their relative activation times.

To explore the possibility that disappearance of Mcm-ChEC signal could be used to distinguish firing of the displaced versus non-displaced populations of Mcm, we first asked whether the pattern of disappearance of Mcm signal recapitulates that of origin firing. In particular, while late origins are inhibited by hydroxyurea (HU), early origins are not; therefore we asked whether this phenomenon was reflected in Mcm-ChEC. We arrested cells in G1 using alpha factor and then released them into medium containing hydroxyurea (HU), taking samples for ChEC analysis at 15, 30, 45 and 90 minutes (**Figure 5A** [↗](#)). HU inhibits ribonucleotide reductase and thereby dramatically slows S phase, making it easier to follow replication kinetics, but the relative order of firing of different origins remains unchanged. In **Figure 5B** [↗](#), we plot the Mcm-ChEC signal at 111 early origins and 101 late origins, with G1 on the x axis and S phase on the Y axis. While we observed little difference in the signal abundance between early and late origins just 15 minutes after release from alpha factor, before origins have had a chance to fire, this difference increases over the time course, with early origins dropping more than late. We conclude that disappearance of Mcm-ChEC signal as cells progress through S phase can be used to assess origin firing.

We then turned our attention to the Mcm signals at the rDNA origin, where the presence of the replication fork block approximately 1.5 kb from the sites of Mcm binding provides a fortuitous means of confirming our interpretation, since loss of Mcm-MNase signal at the origin should appear as a reciprocal increase where the replisome stalls. Mcms in WT cells remained at their initial loading though the duration of experiment (blue box in **Figure 5C** [↗](#), corresponding to position A in **Figure 1C** [↗](#)). Like in wild type, this non-displaced population of Mcm in *sir2* remains relatively constant (blue box); however, in striking contrast, the displaced population of Mcm in *sir2* (orange box, corresponding to positions B and C in **Figure 1C** [↗](#)) shows a continual decrease

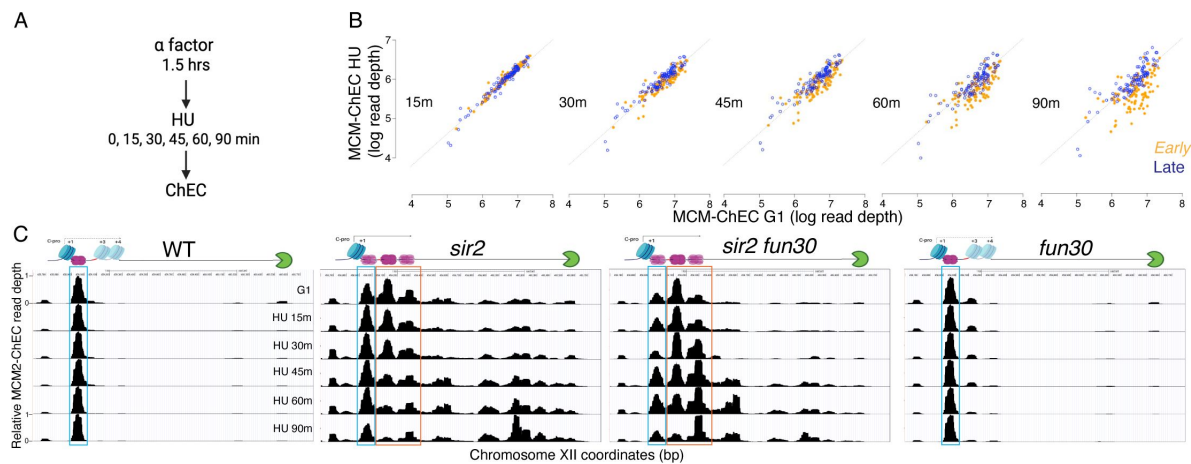


Figure 5.

Disappearance of Mcm-ChEC signal can be used to monitor origin firing.

A. Cells were synchronized in G1 phase using alpha factor for 1.5 hours before being released into media supplemented with hydroxyurea (HU). Cells were harvested at various time points post-release (15-90 minutes) and subjected to Mcm-ChEC analysis. **B.** Decrease in Mcm-ChEC signal in wild type for 111 early (orange) and 101 late (blue) origins. Mcm-ChEC signal was quantified over 200 base pair windows centered on the Mcm binding site within each origin. Each point is plotted according to Mcm-ChEC signal at the time point in question on the y axis and the corresponding signal in G1 on the x axis, thus decrease in signal appears as a drop below the 45° diagonal. **C.** Mcm signal at the rDNA. Mcm-ChEC signal in G1 appears predominantly at the location indicated by the blue rectangle in wild type, whereas it is spread across both the blue and orange rectangles in the absence of Sir2. The displaced Mcm-ChEC signal in *sir2* disappears more rapidly than its non-displaced counterpart, and this effect is suppressed by *fun30*. Decrease of Mcm-ChEC signal at the rARS is accompanied by increased signal to the right.

over the course of the experiment, and there is a concomitant increase in signal in the region upstream of RFB that is absent in wild type. Furthermore, both the decrease in the displaced signal (orange box) and its corresponding increase upstream of RFB were suppressed by deletion of *FUN30*. The changes in the relative abundance of displaced and non-displaced Mcms during the time course, as well as repositioning of the Mcm signal to the region upstream of RFB across different genotypes were highly reproducible (**Figure 5-figure supplement 1** [↗](#)). Taken together, these observations provide strong support for the hypothesis that the displaced population of Mcm in *sir2* cells fires more readily than its non-displaced counterpart and that the increased firing of the displaced Mcms is suppressed by *FUN30* deletion.

It is notable that, although a small fraction of the total G1 signal in wild type arises from the displaced location, this minor signal likewise disappears more rapidly than its non-displaced counterpart (**Figure 5-figure supplement 2** [↗](#)). This suggests that some feature of the immediate chromatin environment may be responsible for the difference in firing times at the two locations, and that this difference is present in both wild type and *sir2*; in other words, the fact that the displaced helicase complex is activated before the non-displaced complex regardless of *SIR2* status is consistent with a model in which *sir2* mutants replicate their rDNA early by virtue of displacing much of their loaded helicases to a region where they are more apt to fire, rather than by altering the chromatin environment more globally to affect all replicative helicases in the nucleolus.

Finally, although deletion of *FUN30* could suppress replication initiation at the rDNA either by inhibiting firing of the active (displaced) Mcm complex, or by preventing its displacement to the “activating location” in the first place, our results make clear that suppression occurs by the former mechanism rather than the latter. Indeed, deletion of *FUN30* actually skews the distribution to favor the displaced location over the non-displaced one, and this skewing can be seen in both *sir2* and *SIR2*. Because, as we have previously shown, rightward displacement of Mcms is accomplished by derepression of C-pro transcription and consequent pushing by RNA PolII, this suggests that deletion of *FUN30* does not act via repression of C-pro transcription. Consistent with both this expectation and a previous report demonstrating that *fun30* mutants are deficient in transcriptional silencing (Neves-Costa et al. 2009 [↗](#)), C-pro RNA levels which were ~2 times higher in *sir2 fun30* cells compared to those in *sir2* cells when adjusted for differences in rDNA size (**Figure 5-figure supplement 3** [↗](#)). Taken together, our findings indicate that *FUN30* deletion specifically hampers the activity of repositioned Mcms when silencing mechanisms are compromised.

Fun30 maintains low nucleosome occupancy near repositioned Mcms

Our findings demonstrate that Mcms, when situated at their loading site adjacent to a high-occupancy +1 nucleosome, exhibit reduced firing propensity. In contrast, when these Mcms are displaced to a neighboring region with lower nucleosome occupancy, their firing propensity significantly increases. This heightened firing tendency is dependent on Fun30 CRE and its remodeling activity. This suggests that the absence of *FUN30* could elevate nucleosome occupancy around the relocated Mcms, consequently inhibiting their ability to fire.

To evaluate this possibility, we examined the effect of *FUN30* deletion on nucleosome occupancy at rDNA origins in WT and *sir2* background using MNase-seq in G1 arrested cells. Chromatin treatment with MNase digests DNA that is not bound by proteins, such as nucleosomes and transcription factors. Nucleosomes, which are wrapped by approximately 150 base pairs of DNA, tend to protect fragments in the 150-200 base pair range, and thus profiling fragments in this size range is widely used to assess nucleosome distribution. Plotting nucleosome midpoints at the rARS in G1 cells (**Figure 6A** [↗](#)) reveals that nucleosome occupancy at the +2 (green box) and +3 (red box) positions is higher in the *sir2 fun30* double mutant than in the *sir2* single mutant. Furthermore, this effect of Fun30 is not limited to the *sir2* background, since nucleosome occupancy is also

modestly increased in *fun30* cells compared to wild type cells. These observations support the notion that it is the relatively nucleosome-free nature of the region into which Mcms are displaced in *sir2* that make the displaced complexes more prone to activation. Furthermore, this suggests that it could be particularly informative to assess nucleosome occupancy specifically adjacent to loaded Mcm complexes, as described below.

It has previously been observed, both with Mcm and with various transcription factors, that the ChEC technique can reveal not only the binding site of the protein to which the MNase is fused, but also, if present, that of the adjacent nucleosome (Zentner et al. 2015 [↗](#); Foss et al. 2019 [↗](#)). This is due to the fact that the entrance and exit sites for DNA wrapped around a nucleosome are in close physical proximity, despite the fact that they are separated by approximately 150 base pairs, and thus the MNase fusion protein adjacent to a nucleosome often cleaves both sites. Mcm-ChEC, therefore, provides a powerful tool for measuring nucleosome occupancy immediately adjacent to the Mcm helicase complex exclusively in those rDNA repeats where those complexes have been loaded. By plotting the midpoints of 150-200 bp fragments released by Mcm2-Mnase, we were able to determine that nucleosome occupancy at both the +2 and +3 nucleosomes, i.e. those adjacent to the displaced Mcm complexes, are elevated in *sir2 fun30* as compared to *sir2* (**Figure 6B** [↗](#)). This indicates that Fun30 is required for maintaining the low nucleosome occupancy in the region into which the preponderance of Mcm complexes have been displaced in *sir2* mutant cells.

Discussion

In this study, we have attempted to shed light on the mechanism by which Sir2 represses replication origins at the rDNA. Sir2's activity has been widely ascribed to the intuitive, but mechanistically imprecise, notion that "closed chromatin" is less accessible to both transcription and replication factors. Consistent with this model, deletion of *SIR2* abolishes heterochromatic compaction and simultaneously increases the ability of methylases and restriction endonucleases to act on those sequences (Gottschling 1992 [↗](#); Singh and Klar 1992 [↗](#); Loo and Rine 1994 [↗](#); Weiss and Simpson 1998 [↗](#); Ansari and Gartenberg 1999 [↗](#); Ravindra et al. 1999 [↗](#)). However, at least in its simplest form, non-specific steric exclusion of DNA metabolic enzymes cannot fully account for the suppression of transcription and replication at the rDNA. Particularly strikingly in its inconsistency with such a model is the fact that the rDNA is the most highly transcribed region of the genome, with transcription by PolI and PolIII accounting for approximately 80% of cellular RNA (Warner 1999 [↗](#); Woolford and Baserga 2013 [↗](#)). These two polymerase complexes are similar in size to the PolII complex, thus size alone is not sufficient to selectively occlude PolII. Furthermore, while deletion of *SIR2* causes a 1.5-fold increase in accessibility of restriction enzymes to heterochromatin at the HML locus, it increases PolIII transcription by >3000 fold (Holland 2002 [↗](#)), thus the magnitude of Sir2's ability to sterically exclude macromolecular complexes from heterochromatin is insufficient to explain its effect on PolIII transcription. Similar measurements in eukaryotic cells found little difference in accessibility of restriction enzymes and micrococcal nuclease between euchromatin and heterochromatin (Chereji et al. 2019 [↗](#)).

In the current report, we argue that Sir2's role in suppressing replication at the rDNA reflects local differences in the distribution of the Mcm replicative helicase complexes rather than more global occlusion of these complexes from the rDNA. In particular, our results suggest that the key difference between wild type and *sir2*, with regard to replication initiation, is the location of the Mcm complex, with the preponderance of that complex in wild type abutting a high-occupancy nucleosome while, in *sir2*, it is mostly in a relatively nucleosome-free region (**Figure 7** [↗](#)).

Furthermore, we show that, even though this nucleosome-free region is less than 200 base pairs from Mcm's original location, the displaced helicase complex is significantly more prone to activation than its non-displaced counterpart. Moreover, the chromatin-remodeling enzyme Fun30 is required to maintain the nucleosome-free character around the displaced Mcm, and its deletion

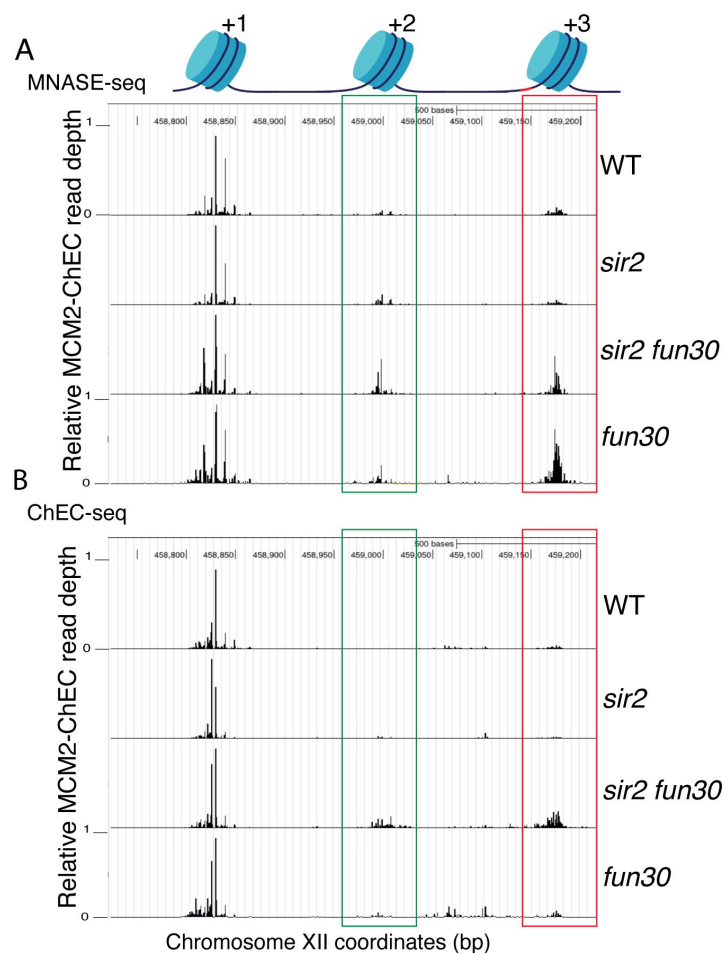


Figure 6.

Nucleosome occupancy assessed by MNase-seq and Mcm-ChEC.

A. Analysis of nucleosome occupancy at rDNA origins using MNase-seq revealed a consistent high occupancy of the +1 nucleosome across all four strain backgrounds, which served as our normalization reference. Occupancy at the +2 (green box) and +3 (red box) positions was increased by deletion of *FUN30* in both *sir2* and *SIR2*. **B.** Analysis of nucleosome occupancy with Mcm-ChEC (see [Figure 1B](#)) reveals nucleosome occupancy in that subset of cells and rDNA repeats in which Mcm is present. Deletion of *FUN30* leads to increased occupancy at the +2 and +3 positions in a *sir2* background.

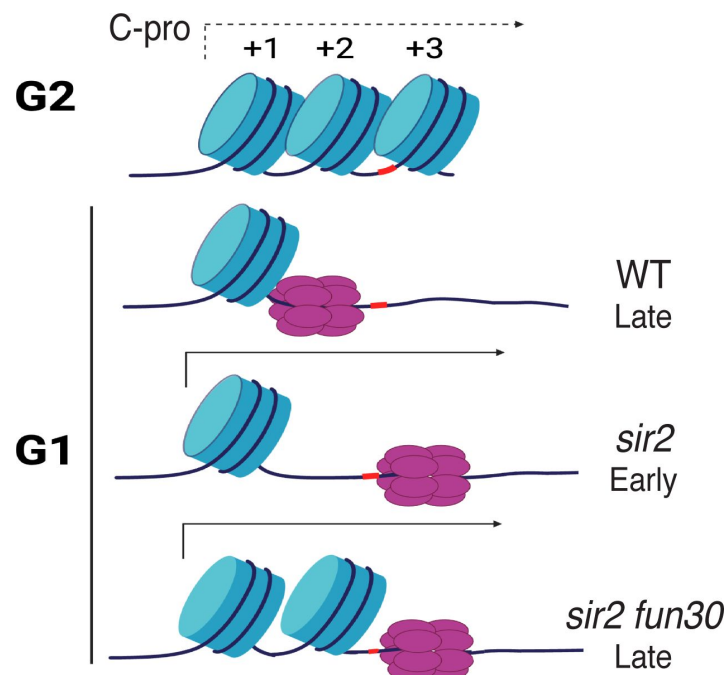


Figure 7.

Model for relationship between Mcm location and replication timing at the rDNA.

Mcm helicase complex (purple ovals) in wild type abuts the +1 nucleosome (blue cylinder) in G1, making it relatively resistant to activation. Deletion of *SIR2* derepresses C-pro transcription (arrow pointing to the right), and RNA PolII pushes Mcm complex to a nucleosome free area, where it is more prone to activation. Deletion of *FUN30* in a *sir2* mutant leads to increased nucleosome occupancy at the +2 position, adjacent to Mcm complex, making this complex resistant to activation. Short red stretch of DNA (e.g. between +2 and +3 nucleosomes in top row) indicates ACS.

both increases nucleosome occupancy and decreases the ability of the displaced Mcm to fire. Finally, as discussed below, we suggest that the increased proclivity of the displaced Mcm to fire is not an anomaly specific to the physiology of *sir2* cells, but is an integral feature of replication of the rDNA during normal S phase.

Our model for the manner in which Sir2 regulates rDNA replication depends critically on the notion that the local nucleosome landscape can influence the activation of Mcm helicase complexes, and *in vitro* experiments suggest that this is, indeed, the case (Azmi et al. 2017 [Azmi et al. 2017](#)): Azmi et al. monitored both licensing and firing *in vitro*, using a 3.8 kb nucleosomal template containing the ARS1 origin. They found that, while licensing was equivalent whether the template had been populated with nucleosomes using Swi/Snf or Rsc remodelers, on the one hand, or Isw1a, on the other, the loaded Mcm complexes in the latter templates were more prone to activation. Furthermore, Mcm complexes on Swi/Snf- or Rsc-remodeled templates could be made more prone to activation if subsequently treated with Isw1a, although the converse was not true. The salient difference between the two types of nucleosomal landscapes, however, was not obvious, as the distance between the Mcm complex and the nearest nucleosome was not notably affected, although it is possible that the composition (e.g. H2A/H2B content) differs. It was also unclear how the nucleosome landscape affected activation of Mcm complexes, though the relevant step in activation was determined to be the formation of the CMG complex with Cdc45 and GINS rather than the prior phosphorylation of the complex by DDK. In summary, while it is clear that the local nucleosome landscape can affect origin activity (Eaton et al. 2010 [Eaton et al. 2010](#); Belsky et al. 2015 [Belsky et al. 2015](#)), the underlying mechanism remains murky.

Regardless of the reason for the increased proclivity of the displaced Mcm to fire, the observation has implications for replication at the rDNA not only in *sir2* mutants but also in wild type. We have previously shown that the rightward displacement of Mcm double hexamers in *sir2* is due to derepression of C-pro transcription (Foss et al. 2019 [Foss et al. 2019](#)), while our current results demonstrate that the displaced Mcm is more prone to activation. Combining these two observations, we suggest that the rDNA replicates earlier in *sir2* than in wild type simply because more of the Mcm complex is in a location that favors its activation; in other words, we favor the view that the region of relatively low nucleosome occupancy to the right of Mcm's normal location is more amenable to Mcm activation, regardless of *SIR2* status. In support of this notion, close analysis of our Mcm-ChEC data reveals that, while Mcm complexes at the displaced location are much rarer in wild type than they are in *sir2*, this particular subpopulation appears equally susceptible to activation in both genotypes. This has the important implication that the manner in which the rDNA origin fires in *sir2* mutants may be simply an exaggerated instance of its normal behavior, and that even in wild type cells, it is predominantly from Mcms that are displaced, likely due to low level of C-pro transcription, that replication initiates.

Classic studies that employed EM and two-dimensional gel electrophoresis showed that only a fraction of rDNA origins, estimated at one in five, are activated in any single cell cycle in yeast (Brewer and Fangman 1988 [Brewer and Fangman 1988](#); Linskens and Huberman 1988 [Linskens and Huberman 1988](#)). These estimates were later corroborated by single molecule analysis, which also observed that the fraction of active origins increases in *sir2* mutants (Pasero et al. 2002 [Pasero et al. 2002](#)). This low fraction of active rDNA origins could be due to insufficient origins licensing, suppression of activation or licensed origins or a combination of the two mechanisms. Our measurements of rDNA licensing showed that MCM are loaded in up to 90% of origins in WT cells and allowed us to conclude that regulation of rDNA replication initiation occurs primarily at the level of origin firing.

Chromatin remodeling enzymes have been proposed to assist DNA replication by modulation of nucleosome phasing in the vicinity of replication origins, but it has proven difficult to assign specific roles to remodelers in genome replication *in vivo* (Cutler et al. 2018 [Cutler et al. 2018](#)). One of the reasons for this difficulty is the redundancy among different remodelers on one hand, and the multiple essential roles CRE play in DNA transactions in addition to replication. For example, *in vitro*

studies have identified four different remodelers that in combination with ORC can phase nucleosome at 320 replication origins (Chacin et al. 2023 [↗](#)). However, a deletion of any single of these remodelers had no discernable effect on nucleosome phasing *in vivo* and a quadruple mutant was required to abolish phasing. While the characterization of this quadruple mutant suggested impaired replication initiation, it was difficult to exclude indirect effects that might have been caused by the effects on transcription. In contrast, the effect of a chromatin remodeler on DNA replication we describe in this work is remarkably specific: we found that the activity of a single remodeler, Fun30, is required to activation not only of a specific origin, but that its action is specific to nucleosomes adjacent to repositioned Mcm complexes at the rDNA. The suppression of rDNA origin activity in sir2 mutants, which contained a large fraction of repositioned Mcms, by *FUN30* deletion is so severe that it reduces the rDNA size ~3-5 fold, from 160 to 35-50 copies.

In conclusion, our findings contribute to a growing body of evidence highlighting the importance of positioning of the Mcm replicative helicase within the local nucleosome environment in determining replication timing (Azmi et al. 2017 [↗](#); Rodriguez et al. 2017 [↗](#)). While it is clear that small differences in this positioning can have significant effects on the timing of DNA replication, it is unlikely that a simple rule exists to directly infer origin timing solely based on the distance between the Mcm complex and the nearest nucleosome. Nonetheless, our results suggest that this parameter could be a crucial piece of the puzzle in understanding why some origins fire early in S phase, while others fire late. Further research is needed to unravel the precise molecular mechanisms governing the interplay between the Mcm complex and nucleosomes and how these dynamics influence replication timing across different genomic loci. Such insights will deepen our understanding of DNA replication regulation in the context of disease states associated with epigenetic drift, such as aging and cancer.

Material and methods

Yeast Strains

Strains used in the study were derived from BY4741 and W303 and are provided in **Supplemental Table 1** [↗](#). Cells were grown in standard yeast peptone 2% dextrose media except for hydroxyurea time course study which employed synthetic media.

Strains containing point mutations in *FUN30* were made using a plasmid that contains CAS9 and gRNA targeting the PAM sites adjacent to the desired mutation sites in *FUN30* as described. Oligos containing *FUN30*-targeting gRNAs were cloned into pML104. pML104 was a gift from John Wyrick (Addgene plasmid # 67638)(Laughery et al. 2015 [↗](#)). The plasmids, which induce double strand breaks within *FUN30*, were co-transformed with DNA blocks containing the repair templates with the desired mutations and altered PAM sites, synthetically synthesized by IDT. *FUN30* DNA was PCR amplified from the resulting transformants and sequenced to verify that the desired mutations have been introduced.

Edu-incorporating yeast strains were derived from W303 and contain four BrdU vectors that express HSV-TK and hENT1 (Viggiani and Aparicio 2006 [↗](#)). Plasmids, containing BrdU cassette obtained from Addgene (Addgene plasmid # 71789-71792), were integrated at four different auxotrophic loci (*HIS3*, *TRP1*, *LEU2*, and *URA3*). Single integration of each BrdU cassette was confirmed via PCR using plasmid-specific primer sets as described (Viggiani and Aparicio 2006 [↗](#)), which assured that the strains contain exactly four copies of the BrdU cassette.

Measurements of rDNA replication timing using S-seq

S-seq experiments were carried out as previously described (Foss et al. 2017 [↗](#)). Cells were grown to log phase in YEPD media, fixed using 70% (vol/vol) ethanol, subjected to proteinase K digestion, and their DNA stained with Sytox Green as described (Foss 2001 [↗](#)). Cells were sorted according to

DNA content on a BD Biosciences FACSaria II cell sorter into G1 and S fractions. DNA from a minimum of 1 e-6 cells from each fraction was isolated using the YeaStar Genomic DNA Kit (Zymo Research). DNA was fragmented by sonication and sequenced.

MNase seq

We carried out MNase-Seq as previously described (Foss 2019). Briefly, cells grown to log phase in rich medium, Yeast Peptone Agar with 2% glucose (YEPD), from an overnight 25 mL culture were synchronized with 3 μ M alpha-factor for 1.5 hrs. at 30 ° C. Arrested cells were crosslinked with 1% formaldehyde for 30 min at room temperature water bath with shaking. Formaldehyde was quenched with 125 mM glycine and cells were centrifuged at 3000 rpm for 5 min. Cells were washed twice with water and resuspended in 1.5 mL Buffer Z (1 M sorbitol, 50 mM Tris-HCl pH 7.4) with 1 mM beta-mercaptoethanol (1.1 μ L of 14.3 M beta-mercaptoethanol diluted 1:10 in Buffer Z) per 25 mL culture. Cells were treated with 100 μ L 20 mg/mL zymolyase at 30 ° C for 20–30 min. Spheroplasts were centrifuged at 5000 rpm for 10 min and resuspended in 5 mL NP buffer (1 M sorbitol, 50 mM NaCl, 10 mM Tris pH 7.4, 5 mM MgCl₂, 1 mM CaCl₂) supplemented with 500 μ M spermidine, 1 mM beta-mercaptoethanol and 0.075% NP-40. Nuclei were aliquoted in tubes with varying concentrations of micrococcal nuclease (Worthington), mixed via tube inversion, and incubated at room temperature for 20 mins. Chromatin digested with 1.9 U– 7.5 U micrococcal nuclease per 1/5th of spheroplasts from a 25 mL culture yielded appropriate mono-, di-, tri-nucleosome protected fragments for next-generation sequencing. Digestion was stopped with freshly made 5x stop buffer (5% SDS, 50 mM EDTA) and proteinase K was added (0.2 mg/ml final concentration) for an overnight incubation at 65 ° C to reverse crosslinking. DNA was extracted with phenol/chloroform and precipitated with ethanol. Micrococcal nuclease digestion was analyzed via gel electrophoresis prior to proceeding to library preparation. Sequencing libraries were prepared as described below for ChEC.

Chromatin Endogenous Cleavage (ChEC)

ChEC-seq was carried out as previously described (Foss et al. 2019 [DOI](#); Foss et al. 2021 [DOI](#)). Briefly, MNase was activated by addition of CaCl₂ to cells that were permeabilized with digitonin. DNA was extracted from cells using phenol and chloroform, precipitated using salt and ethanol and used to construct sequencing libraries without size fractionation. Cells were centrifuged at 1,500 x g for 2 mins, and washed twice in cold Buffer A (15 mM Tris pH 7.5, 80 mM KCl, 0.1 mM EGTA) without additives. Washed cells were carefully resuspended in 570 μ L Buffer A with additives (0.2 mM spermidine, 0.5 mM spermine, 1 mM PMSF, ½ cComplete ULTRA protease inhibitors tablet, Roche, per 5 mL Buffer A) and permeabilized with 0.1% digitonin in 30 ° C water bath for 5 min. Permeabilized cells were cooled at room temperature for 1 min and 1/5th of cells were transferred in a tube with freshly made 2x stop buffer (400 mM NaCl, 20 mM EDTA, 4 mM EGTA)/1% SDS solution for undigested control. Micrococcal nuclease was activated with 5.5 μ L of 200 mM CaCl₂ at various times (5 mins and 10 mins) and the reaction stopped with 2x stop buffer/1% SDS. Once all time points were collected, proteinase K was added to each of the collected time points and incubated at 55 ° C water bath for 30 mins. DNA was extracted using phenol/chloroform and precipitated with ethanol. Micrococcal nuclease digestion was analyzed via gel electrophoresis prior to proceeding to library preparation. Library was prepared as previously described used using total DNA, without any fragment size selection (Foss et al. 2019 [DOI](#); Foss et al. 2021 [DOI](#)).

Quantifying replication using the thymidine analogue Edu

Cells were grown to log phase in rich YEPD medium from a 50 mL culture and synchronized in G1 using 3 μ M alpha factor for 2 hours at 30°C. Using 500 μ g/mL pronase, arrested cells were released into rich media containing 200 mM hydroxyurea and 130 μ M Edu (5-Ethynyl-2'-deoxyuridine) (Biosearch Technologies). Cells were harvested in G1 and after 60 minutes release into HU. DNA was extracted using YeaStar Genomic DNA Kit (Zymo Research) and sonicated. Separate sequencing primers were ligated to each of the samples. One hundred ng of each sample was

pooled together, and subjected to click chemistry reaction in 10 mM Tris pH 7.4, 50 mM NaCl, 20 μ M biotin-TEG azide (Vector Laboratories), 5 mM THPTA (tris-hydroxypropyltriazolylmethylamine) (Vector Laboratories), 1 mM copper sulfate, and 15 mM sodium ascorbate (total volume 200 μ L). Click reaction proceeded for 30 minutes at room temperature prior to DNA ethanol precipitation. Streptavidin magnetic beads (New England Biolabs) were washed 3 times with B&W buffer I (100 mM NaCl, 10 mM Tris-HCl pH 8.0, 1 mM EDTA, 0.05% Tween20, 0.5% SDS) before adding 100 μ L of bead suspension to 180 μ L DNA plus B&W buffer I. After a 30-minute incubation at room temperature while gently inverting tubes, bead bound DNA was washed with 200 μ L B&W buffer I followed by 100 μ L Stringency wash buffer (0.1% SDS, 0.1% SSC buffer) and 200 μ L Wash Buffer II (100 mM NaCl, 10 mM Tris-HCl pH 8.0, 1 mM EDTA, 0.05% Tween 20). DNA was eluted off the beads by denaturation at 98°C for 5 minutes, followed by cooling at 25°C. Edu-labeled DNA was PCR amplified post-pulldown and subjected to sequencing.

Hydroxyurea ChEC time course experiment

Cells grown in synthetic media to logarithmic phase were arrested in G1 using alpha factor for 90 minutes, centrifuged, washed in water to remove alpha factor, and released into media containing 200 mM Hydroxyurea. Cells were harvested in G1 and at indicated time points following release into HU-containing media, chilled at 4°C and analyzed by ChEC as described above.

qRT-PCR analysis of c-pro transcript and rDNA copy number

RNA was extracted from logarithmically growing cells after spheroplasting using an RNA-Easy column. The sequences of the primers used in qRT-PCR for C-pro and *PDA1* mRNA are provided in **Supplemental Table 2** [↗](#). rDNA size was measured by qPCR using DNA that was extracted by YeaStar Genomic DNA Kit (Zymo Research) using the primers listed in **Supplemental Table 2** [↗](#). As an internal standard, qPCR was done in parallel on the DNA extracted from *fof1* strains with 180 and 35 rDNA copies whose rDNA sizes had been verified by Pulsed Field Gel Electrophoresis (Kwan et al. 2023 [↗](#)).

ChEC Southern blots for measuring origin licensing

Cells carrying Mcm2-Mnase were subjected to ChEC as described above, except that the SDS was omitted from the stop buffer to allow subsequent spheroplasting. DNA was extracted from spheroplasts using the YeaStar Genomic DNA Kit (Zymo Research), digested with XmnI, separated on 1.75% agarose gels and analyzed by standard Southern blotting technique. The primers used to make the rDNA and *PIK1* control probe are listed in **Supplemental Table 2** [↗](#).

Acknowledgements

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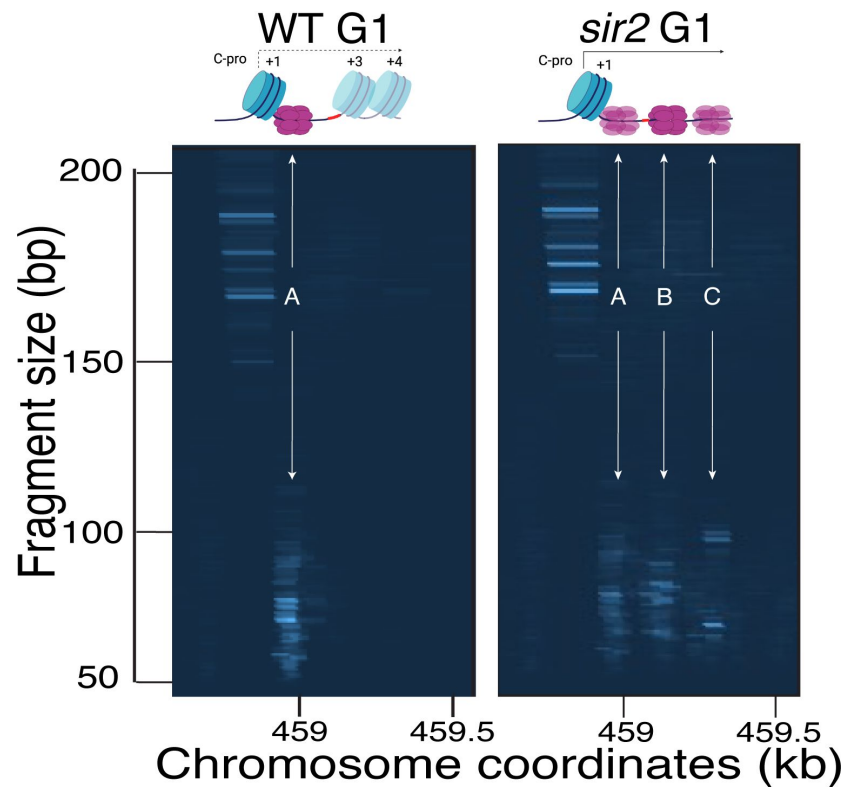


Figure 1-figure supplement 1.

Computer-generated visualization of non-displaced and displaced Mcm complexes, as determined by Mcm-ChEC.

Sequencing reads from G1 cultures processed for Mcm-ChEC for wild type (left) and *sir2* (right) were plotted according to genomic location (x axis) and library insert size (y axis), with read depths indicated by color intensity. Note that these images are not agarose gels. The signal generated from inserts in the 50-100 base pair range (y axis) reflects the Mcm complexes, whereas the signal from inserts in the 150-200 base pair range reflects the +1 nucleosome (see main text). Derepression of C-pro transcription in *sir2* causes RNA PolII to push the Mcm helicase complex from its normal location (arrow labeled "A") to the right, with the most prominent signals arising at the sites indicated by arrows labeled "B" and "C". Note that the presence of multiple Mcm footprints in these composite images does not indicate that the presence of multiple Mcm complexes in any individual repeat.

Figure 2-figure supplement 1.

rDNA size as determined by qPCR.

rDNA size for multiple isolates from each genotype were determined by qPCR, as described (Materials and Methods). Control strains with 35 and 180 copies of the rDNA were generated in *fab1* backgrounds to ensure copy number stability and used as standards. Copy numbers were 160 ± 7 , 156 ± 7 , 38 ± 3 and 147 ± 7 for WT, *sir2*, *sir2fun30* and *fun30*, respectively (t-test $p < 0.001$ for *sir2* vs *sir2 fun30*).

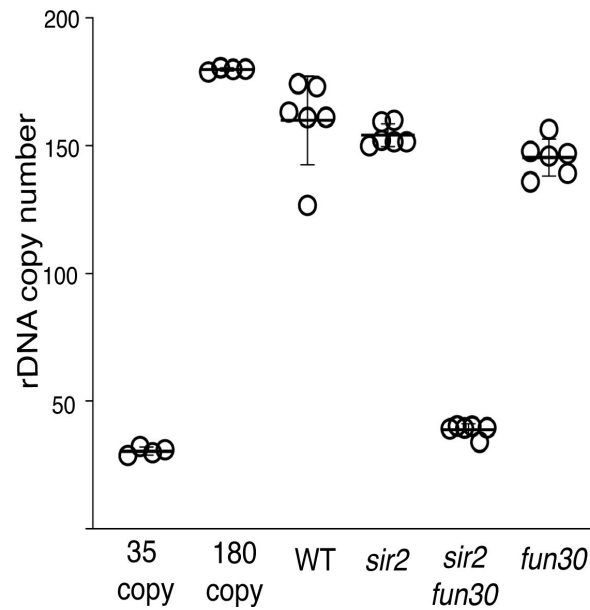
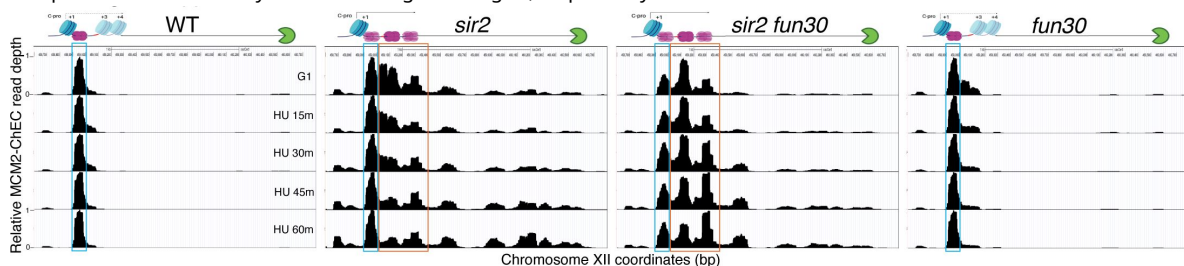


Figure 5-figure supplement 1.

Deletion of *FUN30* suppresses activation of the displaced Mcm complex at the rDNA (replica of results in Figure 5C [↗](#)).

Each panel shows MCM-ChEC data for cultures that were arrested in G1 and then released into medium containing 200 mM HU. Activation of Mcm complexes is inferred from disappearance of MCM-ChEC signal. Non-displaced and displaced Mcm complexes are indicated by blue and orange rectangles, respectively.



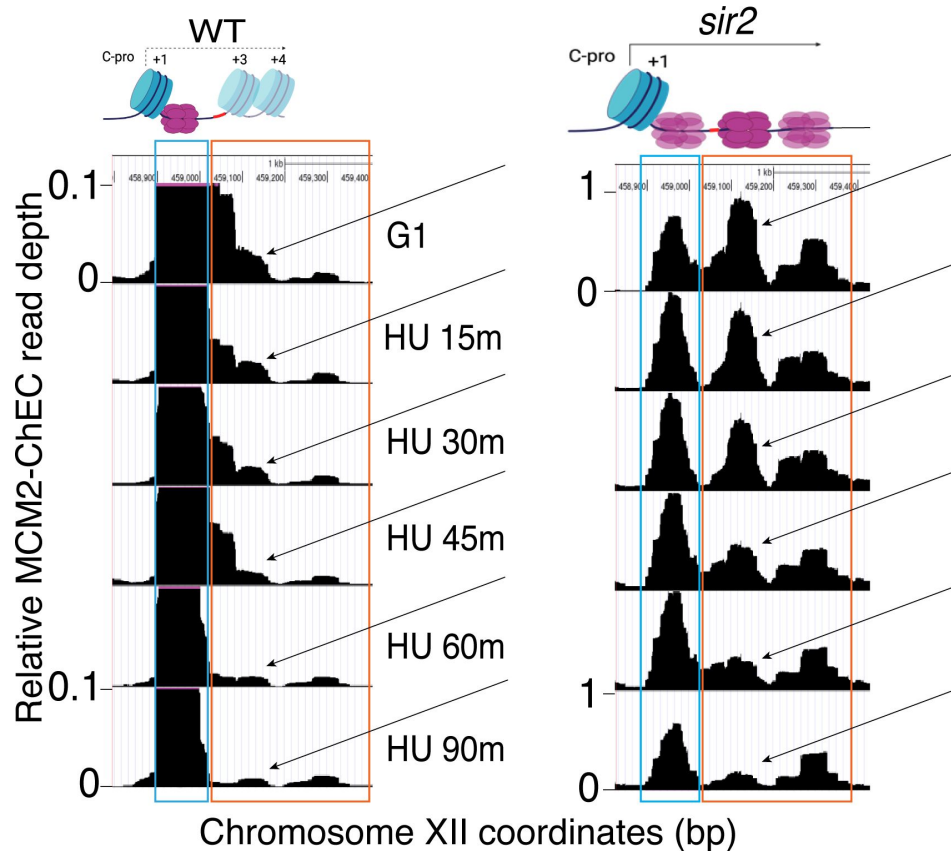


Figure 5-figure supplement 2.

Displaced Mcm complex is activated early in wild type.

Enlarged region from [figure 5C](#) shows that, although only a small proportion of Mcm complexes are displaced in WT, this displaced population, indicated by arrows, is activated early, as is the case with in *sir2*.

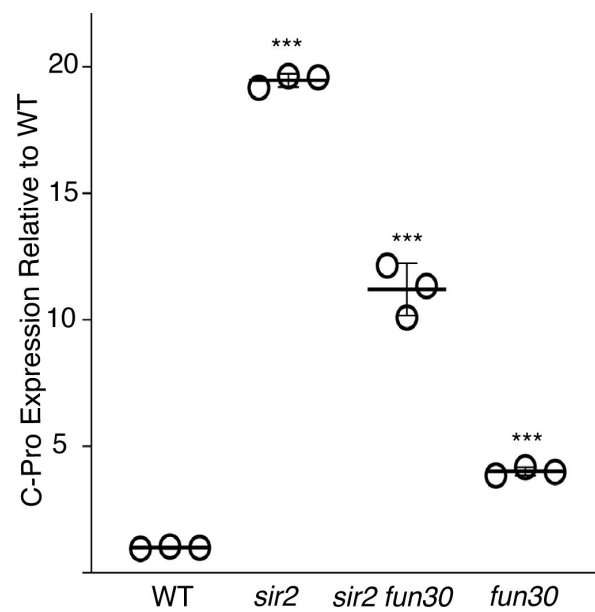


Figure 5-figure supplement 3.

C-pro transcript levels.

C-pro transcript levels in G1-arrested cells were measured in triplicate using qPCR and expressed relative to WT. *** denotes $p < 0.001$ for comparison with WT using t-test.

Strain	Genotype
W1588-4c*	<i>MATa, ade2-1, Can1-100, his3-11,15, leu2-3,112, trp1-1, ura3-1, RAD5+</i> , W303 background
16535	<i>MATa, his3, leu2, met15, ura3, hmlα::HYG</i>
16668	<i>MATa, his3, leu2, met15, ura3, hmlα::HYG, sir2::KanMX</i>
16673	<i>MATa, his3, leu2, met15, ura3, hmlα::HYG, isw2::NAT, sir2::LEU</i>
16723	<i>MATa, his3, leu2, met15, ura3, irc5::KanMX, sir2::LEU</i>
16724	<i>MATa, his3, leu2, met15, ura3, isw1::KanMX, sir2::LEU</i>
16725	<i>MATa, his3, leu2, met15, ura3, chd1::KanMX, sir2::LEU</i>
16728	<i>MATa, his3, leu2, met15, ura3, swr1::KanMX, sir2::HYG</i>
16729	<i>MATa, his3, leu2, met15, ura3, nhp10::KanMX, sir2::LEU</i>
16747	<i>MATa, his3, leu2, met15, ura3, hmlα::HYG</i> , with <i>MCM2</i> MNase 3xFlag Tag with KanMx (161 rDNA copies)
16769	<i>MATa, his3, leu2, met15, ura3, hmlα::HYG, sir2::LEU</i> , with <i>MCM2</i> MNase 3xFlag Tag with KanMx (152 rDNA copies)
16909	<i>MATa, his3, leu2, met15, ura3, fun30::KanMX, sir2::LEU, hmlα::HYG</i>
17059	<i>MATa, his3, leu2, met15, ura3, htl1::NAT, sir2::LEU</i>
17113	<i>MATa, his3, leu2, met15, ura3, hmlα::HYG, fun30S20AS28A, sir2::LEU</i> (164 rDNA copies)
17114	<i>MATa, his3, leu2, met15, ura3, hmlα::HYG, fun30S20AS28A, sir2::LEU</i> (166 rDNA copies)
17248	<i>MATa, his3, leu2, met15, ura3, hmlα::HYG, fun30::URA</i>
17256	<i>MATa, his3, leu2, met15, ura3, hmlα::HYG, fun30::NAT</i> , with <i>MCM2</i> MNase 3xFlag Tag with KanMx (146 rDNA copies)
17257	<i>MATa, his3, leu2, met15, ura3, hmlα::HYG, sir2::LEU, fun30::NAT</i> , with <i>MCM2</i> MNase 3xFlag Tag with KanMx (39 rDNA copies)
17265	<i>MATa, ade2-1, Can1-100, his3-11,15, leu2-3,112, trp1-1, ura3-1, RAD5+</i> , with p306-BrdU-Inc, p403-BrdU-Inc, p405-BrdU-Inc, p404-BrdU-Inc, <i>hmlα::HYG</i> , W303 background
17271	<i>MATa, ade2-1, Can1-100, his3-11,15, leu2-3,112, trp1-1, ura3-1, RAD5+</i> , with p306-BrdU-Inc, p403-BrdU-Inc, p405-BrdU-Inc, p404-BrdU-Inc, <i>hmlα::HYG, sir2::KanMX</i> , W303 background
17279	<i>MATa, ade2-1, Can1-100, his3-11,15, leu2-3,112, trp1-1, ura3-1, RAD5+</i> , with p306-BrdU-Inc, p403-BrdU-Inc, p405-BrdU-Inc, p404-BrdU-Inc, <i>hmlα::HYG, sir2::KanMX, fun30::NAT</i> , W303 background
17281	<i>MATa, ade2-1, Can1-100, his3-11,15, leu2-3,112, trp1-1, ura3-1, RAD5+</i> , with p306-BrdU-Inc, p403-BrdU-Inc, p405-BrdU-Inc, p404-BrdU-Inc, <i>hmlα::HYG, fun30::NAT</i> , W303 background
17345	<i>MATa, his3, leu2, met15, ura3, hmlα::HYG, fun30K603R, sir2::LEU</i>
17346	<i>MATa, his3, leu2, met15, ura3, hmlα::HYG, fun30K603R, sir2::LEU</i>
EK342**	<i>MATa, his3, leu2, met15, ura3, fob1::NAT</i> ; BY rDNA (35 rDNA copies)
EK68**	<i>MATa</i> , Prototroph, <i>fob1::NAT</i> ; BY rDNA (180 rDNA copies)

*Strain from Toshi Tsukiyama (Rodriguez et al. 2017)

** Strain from Bonny Brewer and M.K. Raghuraman (Kwan et al. 2023)

Table S1

Yeast strains

Primers	Sequence (5' to 3')	Comments
5P_c_pro_qPCR	CCATCTATGTCTTCCACACC	5' primer for C-pro qRT-PCR
3P_c_pro_qPCR	GTCATTTACAAGAGGTAGGTCG	3' primer for C-pro qRT-PCR
PDA1_1_F	GCTGCTTCATTCAAACGCCA	5' primer for <i>PDA1</i> control for C-pro qRT-PCR
PDA1_1_R	CTGGTGGGAGTGCGAAGAAC	3' primer for <i>PDA1</i> control for C-pro qRT-PCR
Tor2_5p_qPCR	GCACAAAGACCACAAAGTCG	5' primer for <i>TOR2</i> control for rDNA size qPCR
Tor2_3p_qPCR	GGGGATAGAGAACTAACAAAAGCA	3' primer for <i>TOR2</i> control for rDNA size qPCR
ARS1200_qPCR Forward	GAAATACGCTTCAGAGACCC	5' primer for ARS1200 for rDNA size qPCR
ARS1200_qPCR Reverse	CGGAAATGGCTGATTTTGATTG	3' primer for ARS1200 for rDNA size qPCR
rARS_southern_5prime1	ACTTCAATTTAAGCTATTCTCC	5' primer for ARS1200 probe for Southern Blot
rARS_3Primer_southern_1	TGCAAGTAGCATATATTCTTG	3' primer for ARS1200 probe for Southern Blot
Pik1_F	GATTCAGCGTCCACAGCTC	5' primer for <i>PIK1</i> control probe for Southern Blot
Pik1_B	GGTAACCAAATTTATCCTGCACC	3' primer for <i>PIK1</i> control probe for Southern Blot

Table S2

Primers used

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Reviewer #1 (Public Review):

Summary:

This paper presents a mechanistic study of rDNA origin regulation in yeast by SIR2. Each of the ~180 tandemly repeated rDNA gene copies contains a potential replication origin. Early-efficient initiation of these origins is suppressed by Sir2, reducing competition with origins distributed throughout the genome for rate-limiting initiation factors. Previous studies by these authors showed that SIR2 deletion advances replication timing of rDNA origins by a complex mechanism of transcriptional de-repression of a local PolII promoter causing licensed origin proteins (MCM complexes) to re-localize (slide along the DNA) to a different (and altered) chromatin environment. In this study, they identify a chromatin remodeler, FUN30, that suppresses the sir2Δ effect, and remarkably, results in a contraction of the rDNA to about one-quarter its normal length/number of repeats, implicating replication defects of the rDNA. Through examination of replication timing, MCM occupancy and nucleosome occupancy on the chromatin in sir2, fun30, and double mutants, they propose a model where nucleosome position relative to the licensed origin (MCM complexes) intrinsically determines origin timing/efficiency. While their interpretations of the data are largely reasonable and can be interpreted to support their model, a key weakness is the connection between MCM ChEC signal disappearance and origin firing. While the cyclical chromatin association-dissociation of MCM proteins with potential origin sequences may be generally interpreted as licensing followed by firing, dissociation may also result from passive replication and as shown here, displacement by transcription and/or chromatin remodeling. Moreover, linking its disappearance from chromatin in the ChEC method with such precise resolution needs to be validated against an independent method to determine the initiation site(s). Differences in rDNA copy number and relative transcription levels also are not directly accounted for, obscuring a clearer interpretation of the results. Nevertheless, this paper makes a valuable advance with the finding of Fun30 involvement, which substantially reduces rDNA repeat number in sir2Δ background. The model they develop is compelling and I am inclined to agree, but I think the evidence on this specific point is purely correlative and a better method is needed to address the initiation site question. The authors deserve credit for their efforts to elucidate our obscure understanding of the intricacies of chromatin regulation. At a minimum, I suggest their conclusions on these points of concern should be softened and caveats discussed. Statistical analysis is lacking for some claims.

Strengths are the identification of FUN30 as suppressor, examination of specific mutants of FUN30 to distinguish likely functional involvement. Use of multiple methods to analyze replication and protein occupancies on chromatin. Development of a coherent model.

Weaknesses are failure to address copy number as a variable; insufficient validation of ChEC method relationship to exact initiation locus; lack of statistical analysis in some cases.

Additional background and discussion for public review:

This paper broadly addresses the mechanism(s) that regulate replication origin firing in different chromatin contexts. The rDNA origin is present in each of ~180 tandem repeats of the rDNA sequence, representing a high potential origin density per length of DNA (9.1kb repeat unit). However, the average origin efficiency of rDNA origins is relatively low (~20% in wild-type cells), which reduces the replication load on the overall genome by reducing competition with origins throughout the genome for limiting replication initiation factors. Deletion of histone deacetylase SIR2, which silences PolII transcription within the rDNA, results in increased early activation of the rDNA origins (and reduced rate of overall genome replication). Previous work by the authors showed that MCM complexes loaded onto the rDNA origins (origin licensing) were laterally displaced (sliding) along the rDNA, away from a well-positioned nucleosome on one side. The authors' major hypothesis throughout this work is that the new MCM location(s) are intrinsically more efficient configurations for origin firing. The authors identify a chromatin remodeling enzyme, FUN30, whose deletion appears to suppress the earlier activation of rDNA origins in *sir2Δ* cells. Indeed, it appears that the reduction of rDNA origin activity in *sir2Δ fun30Δ* cells is severe enough to result in a substantial reduction in the rDNA array repeat length (number of repeats); the reduced rDNA length presumably facilitates its more stable replication and maintenance.

Analysis of replication by 2D gels is marginally convincing, using 2D gels for this purpose is very challenging and tricky to quantify. The more quantitative analysis by EdU incorporation is more convincing of the suppression of the earlier replication caused by SIR2 deletion.

To address the mechanism of suppression, they analyze MCM positioning using ChEC, which in G1 cells shows partial displacement of MCM from normal position A to positions B and C in *sir2Δ* cells and similar but more complete displacement away from A to positions B and C in *sir2fun30Δ* cells. During S-phase in the presence of hydroxyurea, which slows replication progression considerably (and blocks later origin firing) MCM signals redistribute, which is interpreted to represent origin firing and bidirectional movement of MCMs (only one direction is shown), some of which accumulate near the replication fork barrier, consistent with their interpretation. They observe that MCMs displaced (in G1) to sites B or C in *sir2Δ* cells, disappear more rapidly during S-phase, whereas the similar dynamic is not observed in *sir2Δfun30Δ*. This is the main basis for their conclusion that the B and C sites are more permissive than A. While this may be the simplest interpretation, there are limitations with this assay that undermine a rigorous conclusion (additional points below). The main problem is that we know the MCM complexes are mobile so disappearance may reflect displacement by other means including transcription which is high in the *sir2Δ* background. Indeed, the double mutant has greater level of transcription per repeat unit which might explain more displaced from A in G1. Thus, displacement might not always represent origin firing. Because the *sir2* background profoundly changes transcription, and the double mutant has a much smaller array length associated with higher transcription, how can we rule out greater accessibility at site A, for example in *sir2Δ*, leading to more firing, which is suppressed in *sir2 fun30* due to greater MCM displacement away from A?

I think the critical missing data to solidly support their conclusions is a definitive determination of the site(s) of initiation using a more direct method, such as strand specific sequencing of EdU or nascent strand analysis. More direct comparisons of the strains with

lower copy number to rule out this facet. As discussed in detail below, copy number reduction is known to suppress at least part of the *sir2Δ* effect so this looms over the interpretations. I think they are probably correct in their overall model based on the simplest interpretation of the data but I think it remains to be rigorously established. I think they should soften their conclusions in this respect.

<https://doi.org/10.7554/eLife.97438.1.sa2>

Reviewer #2 (Public Review):

Summary:

In this manuscript, the authors follow up on their previous work showing that in the absence of the Sir2 deacetylase the MCM replicative helicase at the rDNA spacer region is repositioned to a region of low nucleosome occupancy. Here they show that the repositioned displaced MCMs have increased firing propensity relative to non-displaced MCMs. In addition, they show that activation of the repositioned MCMs and low nucleosome occupancy in the adjacent region depend on the chromatin remodeling activity of Fun30.

Strengths:

The paper provides new information on the role of a conserved chromatin remodeling protein in the regulation of origin firing and in addition provides evidence that not all loaded MCMs fire and that origin firing is regulated at a step downstream of MCM loading.

Weaknesses:

The relationship between the author's results and prior work on the role of Sir2 (and Fob1) in regulation of rDNA recombination and copy number maintenance is not explored, making it difficult to place the results in a broader context. Sir2 has previously been shown to be recruited by Fob1, which is also required for DSB formation and recombination-mediated changes in rDNA copy number. Are the changes that the authors observe specifically in *fun30 sir2* cells related to this pathway? Is Fob1 required for the reduced rDNA copy number in *fun30 sir2* double mutant cells?

<https://doi.org/10.7554/eLife.97438.1.sa1>

Reviewer #3 (Public Review):

Summary:

Heterochromatin is characterized by low transcription activity and late replication timing, both dependent on the NAD-dependent protein deacetylase Sir2, the founding member of the sirtuins. This manuscript addresses the mechanism by which Sir2 delays replication timing at the rDNA in budding yeast. Previous work from the same laboratory (Foss et al. *PLoS Genetics* 15, e1008138) showed that Sir2 represses transcription-dependent displacement of the Mcm helicase in the rDNA. In this manuscript, the authors show convincingly that the repositioned Mcms fire earlier and that this early firing partly depends on the ATPase activity of the nucleosome remodeler Fun30. Using read-depth analysis of sorted G1/S cells, *fun30* was the only chromatin remodeler mutant that somewhat delayed replication timing in *sir2* mutants, while *nhp10*, *chd1*, *isw1*, *htl1*, *swr1*, *isw2*, and *irc5* had no effect. The conclusion was corroborated with orthogonal assays including two-dimensional gel electrophoresis and analysis of EdU incorporation at early origins. Using an insightful analysis with an Mcm-MNase fusion (Mcm-ChEC), the authors show that the repositioned Mcms in *sir2* mutants fire earlier than the Mcm at the normal position in wild type. This early firing at the repositioned

Mcm is partially suppressed by Fun30. In addition, the authors show Fun30 affects nucleosome occupancy at the sites of the repositioned Mcm, providing a plausible mechanism for the effect of Fun30 on Mcm firing at that position. However, the results from the MNase-seq and ChEC-seq assays are not fully congruent for the fun30 single mutant. Overall, the results support the conclusions providing a much better mechanistic understanding how Sir2 affects replication timing at rDNA,

Strengths

- (1) The data clearly show that the repositioned Mcm helicase fires earlier than the Mcm in the wild type position.
- (2) The study identifies a specific role for Fun30 in replication timing and an effect on nucleosome occupancy around the newly positioned Mcm helicase in sir2 cells.

Weaknesses

- (1) It is unclear which strains were used in each experiment.
- (2) The relevance of the fun30 phospho-site mutant (S20AS28A) is unclear.
- (3) For some experiments (Figs. 3, 4, 6) it is unclear whether the data are reproducible and the differences significant. Information about the number of independent experiments and quantitation is lacking. This affects the interpretation, as fun30 seems to affect the +3 nucleosome much more than let on in the description.

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Author response:

eLife assessment

This study is a detailed investigation of how chromatin structure influences replication origin function in yeast ribosomal DNA, with focus on the role of the histone deacetylase Sir2 and the chromatin remodeler Fun30. Convincing evidence shows that Sir2 does not affect origin licensing but rather affects local transcription and nucleosome positioning which correlates with increased origin firing. However, the evidence remains incomplete as the methods employed do not rigorously establish a key aspect of the mechanism, fully address some alternative models, or sufficiently relate to prior results. Overall, this is a valuable advance for the field that could be improved to establish a more robust paradigm.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

This paper presents a mechanistic study of rDNA origin regulation in yeast by SIR2. Each of the ~180 tandemly repeated rDNA gene copies contains a potential replication origin. Early-efficient initiation of these origins is suppressed by Sir2, reducing competition with origins distributed throughout the genome for rate-limiting initiation factors. Previous studies by these authors showed that SIR2 deletion advances replication timing of rDNA origins by a complex mechanism of transcriptional de-repression of a local PolII promoter causing licensed origin proteins (MCM complexes) to re-localize (slide along the DNA) to a different (and altered) chromatin environment. In this study, they identify a chromatin remodeler, FUN30, that suppresses the sir2Δ effect, and remarkably, results in a contraction of the

*rDNA to about one-quarter its normal length/number of repeats, implicating replication defects of the rDNA. Through examination of replication timing, MCM occupancy and nucleosome occupancy on the chromatin in *sir2*, *fun30*, and double mutants, they propose a model where nucleosome position relative to the licensed origin (MCM complexes) intrinsically determines origin timing/efficiency. While their interpretations of the data are largely reasonable and can be interpreted to support their model, a key weakness is the connection between Mcm ChEC signal disappearance and origin firing. While the cyclical chromatin association-dissociation of MCM proteins with potential origin sequences may be generally interpreted as licensing followed by firing, dissociation may also result from passive replication and as shown here, displacement by transcription and/or chromatin remodeling.*

While it is true that both transcription and passive replication can cause the signal of MCM-ChEC to disappear, neither can cause selective disappearance of the displaced complex without affecting the non-displaced complex. Indeed, in the case of transcription, RNA polymerase transcribing C-pro would have to first dislodge the normally positioned MCM complex before even reaching the displaced complex. Furthermore, deletion of FUN30 leads to both more C-pro transcription and less disappearance of the displaced MCM complex. It is important to keep in mind that this cannot somehow reflect continuous replenishment of displaced MCMs with newly loaded MCMs, since the cells are in S phase and licensing is restricted to G1.

Moreover, linking its disappearance from chromatin in the ChEC method with such precise resolution needs to be validated against an independent method to determine the initiation site(s). Differences in rDNA copy number and relative transcription levels also are not directly accounted for, obscuring a clearer interpretation of the results.

Copy number reduction of the magnitude caused by deletion of SIR2 and FUN30 does not suppress the sir2D effect (i.e. early replication of the rDNA), but rather exacerbates it. In particular, deletion of SIR2 and FUN30 causes the rDNA to shrink to approximately 35 copies. Kwan et al., 2023 (PMID: 36842087) have shown that reduction of rDNA copy number to 35 causes a dramatic acceleration of rDNA replication in a SIR2 strain. Thus, the effect of rDNA size on replication timing reinforces our conclusion that deletion of FUN30 suppresses rDNA replication.

However, to address this concern directly, in the revision we will include 2 D gels in *fob1* strains with equal number of repeats that allows to conclude that the effect of FUN30 deletion in suppressing rDNA origin firing is independent of either rDNA size or FOB1. The figure of the critical 2 D gels is shown below in the reply to reviewer 2.

*Nevertheless, this paper makes a valuable advance with the finding of Fun30 involvement, which substantially reduces rDNA repeat number in *sir2Δ* background. The model they develop is compelling and I am inclined to agree, but I think the evidence on this specific point is purely correlative and a better method is needed to address the initiation site question. The authors deserve credit for their efforts to elucidate our obscure understanding of the intricacies of chromatin regulation. At a minimum, I suggest their conclusions on these points of concern should be softened and caveats discussed. Statistical analysis is lacking for some claims.*

Strengths are the identification of FUN30 as suppressor, examination of specific mutants of FUN30 to distinguish likely functional involvement. Use of multiple methods to analyze replication and protein occupancies on chromatin. Development of a coherent model.

Weaknesses are failure to address copy number as a variable; insufficient validation of ChEC method relationship to exact initiation locus; lack of statistical analysis in some cases.

The two potential initiation sites that one would monitor (non-displaced and displaced) are separated by less than 150 base pairs, and other techniques simply do not have the resolution necessary to distinguish such differences. Furthermore, as we suggest in the manuscript, our results are consistent with a model in which it is only the displaced MCM complex that is activated, whether in *sir2* or WT. If no genotype-dependent difference in initiation sites is even expected, it would be hard to interpret even the most precise replication-based assays. However, the reviewer is correct that this is a novel technique and that confirmation with a well-established technique is comforting, therefore we are performing ChIP experiments to corroborate, to the extent possible, the conclusions that we reached with ChEC.

We appreciate the reviewer pointing out that some statistical analyses were lacking, and we will correct this in a revised manuscript.

Additional background and discussion for public review:

*This paper broadly addresses the mechanism(s) that regulate replication origin firing in different chromatin contexts. The rDNA origin is present in each of ~180 tandem repeats of the rDNA sequence, representing a high potential origin density per length of DNA (9.1kb repeat unit). However, the average origin efficiency of rDNA origins is relatively low (~20% in wild-type cells), which reduces the replication load on the overall genome by reducing competition with origins throughout the genome for limiting replication initiation factors. Deletion of histone deacetylase SIR2, which silences PolII transcription within the rDNA, results in increased early activation of the rDNA origins (and reduced rate of overall genome replication). Previous work by the authors showed that MCM complexes loaded onto the rDNA origins (origin licensing) were laterally displaced (sliding) along the rDNA, away from a well-positioned nucleosome on one side. The authors' major hypothesis throughout this work is that the new MCM location(s) are intrinsically more efficient configurations for origin firing. The authors identify a chromatin remodeling enzyme, FUN30, whose deletion appears to suppress the earlier activation of rDNA origins in *sir2Δ* cells. Indeed, it appears that the reduction of rDNA origin activity in *sir2Δ fun30Δ* cells is severe enough to result in a substantial reduction in the rDNA array repeat length (number of repeats); the reduced rDNA length presumably facilitates its more stable replication and maintenance.*

Analysis of replication by 2D gels is marginally convincing, using 2D gels for this purpose is very challenging and tricky to quantify. The more quantitative analysis by EdU incorporation is more convincing of the suppression of the earlier replication caused by SIR2 deletion.

*To address the mechanism of suppression, they analyze MCM positioning using ChEC, which in G1 cells shows partial displacement of MCM from normal position A to positions B and C in *sir2Δ* cells and similar but more complete displacement away from A to positions B and C in *sir2fun30Δ* cells. During S-phase in the presence of hydroxyurea, which slows replication progression considerably (and blocks later origin firing) MCM signals redistribute, which is interpreted to represent origin firing and bidirectional movement of MCMs (only one direction is shown), some of which accumulate near the replication fork barrier, consistent with their interpretation. They observe that MCMs displaced (in G1) to sites B or C in *sir2Δ* cells, disappear more rapidly during S-phase, whereas the similar dynamic is not observed in *sir2Δfun30Δ*. This is the main basis for their conclusion that the B and C sites are more permissive than A. While this may be the simplest interpretation, there are limitations with this assay that undermine a rigorous*

conclusion (additional points below). The main problem is that we know the MCM complexes are mobile so disappearance may reflect displacement by other means including transcription which is high in the *sir2Δ* background. Indeed, the double mutant has greater level of transcription per repeat unit which might explain more displacement from A in G1. Thus, displacement might not always represent origin firing. Because the *sir2* background profoundly changes transcription, and the double mutant has a much smaller array length associated with higher transcription, how can we rule out greater accessibility at site A, for example in *sir2Δ*, leading to more firing, which is suppressed in *sir2 fun30* due to greater MCM displacement away from A?

I think the critical missing data to solidly support their conclusions is a definitive determination of the site(s) of initiation using a more direct method, such as strand specific sequencing of EdU or nascent strand analysis. More direct comparisons of the strains with lower copy number to rule out this facet. As discussed in detail below, copy number reduction is known to suppress at least part of the *sir2Δ* effect so this looms over the interpretations. I think they are probably correct in their overall model based on the simplest interpretation of the data but I think it remains to be rigorously established. I think they should soften their conclusions in this respect.

Reviewer #2 (Public Review):

Summary:

In this manuscript, the authors follow up on their previous work showing that in the absence of the Sir2 deacetylase the MCM replicative helicase at the rDNA spacer region is repositioned to a region of low nucleosome occupancy. Here they show that the repositioned displaced MCMs have increased firing propensity relative to non-displaced MCMs. In addition, they show that activation of the repositioned MCMs and low nucleosome occupancy in the adjacent region depend on the chromatin remodeling activity of Fun30.

Strengths:

The paper provides new information on the role of a conserved chromatin remodeling protein in the regulation of origin firing and in addition provides evidence that not all loaded MCMs fire and that origin firing is regulated at a step downstream of MCM loading.

Weaknesses:

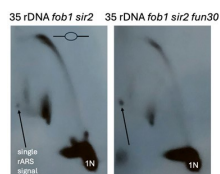
*The relationship between the author's results and prior work on the role of Sir2 (and Fob1) in regulation of rDNA recombination and copy number maintenance is not explored, making it difficult to place the results in a broader context. Sir2 has previously been shown to be recruited by Fob1, which is also required for DSB formation and recombination-mediated changes in rDNA copy number. Are the changes that the authors observe specifically in *fun30 sir2* cells related to this pathway? Is Fob1 required for the reduced rDNA copy number in *fun30 sir2* double mutant cells?*

Strains lacking *SIR2* have unstable rDNA size, and *FOB1* deletion stabilizes rDNA size in *sir2* background. Likewise, *FOB1* deletion influences the kinetics rDNA size reduction in *sir2 fun30* cells. However, the main effect of *Fun30* in *sir2* cells we were interested in, suppression of rDNA replication, is preserved in *fob1* background, arguing that the observed effect is independent of *Fob1* (see figure below). Given that the main focus of the paper is regulation of rDNA origins activity and that these changes were independent of *Fob1*, we had elected not to include these results in the original manuscript but will gladly include them in the revision.

Besides refuting the possible role of Fob1 in the *FUN30*-mediated activation of rDNA origin firing in *sir2* cells, the use of *fob1* background enabled us compare the activation of rDNA origins in the *sir2* and *sir2 fun30* strains with equally short rDNA size. The 2-D gels demonstrate a dramatic suppression of rDNA origin activity upon deletion of *FUN30* in the *sir2 fob1* strains with 35 rDNA copies.

Author response image 1.

The deletion of *FUN30* diminishes the replication bubble signal in a *fob1 sir2* strain with 35 rDNA copies by more than tenfold. The single rARS signal, marked with the arrow, originates from the rightmost rDNA repeat. This specific rightmost rDNA *NheI* fragment is approximately 25 kb in size, distinctly larger than the 4.7 kb *NheI* 1N rARS-containing fragments that originate from the internal rDNA repeats.



Reviewer #3 (Public Review):

Summary:

Heterochromatin is characterized by low transcription activity and late replication timing, both dependent on the NAD-dependent protein deacetylase Sir2, the founding member of the sirtuins. This manuscript addresses the mechanism by which Sir2 delays replication timing at the rDNA in budding yeast. Previous work from the same laboratory (Foss et al. PLoS Genetics 15, e1008138) showed that Sir2 represses transcription-dependent displacement of the Mcm helicase in the rDNA. In this manuscript, the authors show convincingly that the repositioned Mcms fire earlier and that this early firing partly depends on the ATPase activity of the nucleosome remodeler Fun30. Using read-depth analysis of sorted G1/S cells, fun30 was the only chromatin remodeler mutant that somewhat delayed replication timing in sir2 mutants, while nhp10, chd1, isw1, htl1, swr1, isw2, and irc5 had not effect. The conclusion was corroborated with orthogonal assays including two-dimensional gel electrophoresis and analysis of EdU incorporation at early origins. Using an insightful analysis with an Mcm-MNase fusion (Mcm-ChEC), the authors show that the repositioned Mcms in sir2 mutants fire earlier than the Mcm at the normal position in wild type. This early firing at the repositioned Mcms is partially suppressed by Fun30. In addition, the authors show Fun30 affects nucleosome occupancy at the sites of the repositioned Mcm, providing a plausible mechanism for the effect of Fun30 on Mcm firing at that position. However, the results from the MNase-seq and ChEC-seq assays are not fully congruent for the fun30 single mutant. Overall, the results support the conclusions providing a much better mechanistic understanding how Sir2 affects replication timing at rDNA.

The reason that the results for the *fun30* single mutant appear incongruent, with a larger signal of the +2 nucleosome in the MNase-seq plot but a negligible signal in the ChEC-seq plot is the paucity of displaced Mcm in the *fun30* single mutant. Given the relative absence of displaced MCMs, the MCM-MNase fusion protein can't "light up" the +2 nucleosome. We will comment on this in the revision to clarify this.

Strengths

(1) The data clearly show that the repositioned Mcm helicase fires earlier than the Mcm in the wild type position.

(2) The study identifies a specific role for Fun30 in replication timing and an effect on nucleosome occupancy around the newly positioned Mcm helicase in sir2 cells.

Weaknesses

(1) It is unclear which strains were used in each experiment.

(2) The relevance of the fun30 phospho-site mutant (S20AS28A) is unclear.

(3) For some experiments (Figs. 3, 4, 6) it is unclear whether the data are reproducible and the differences significant. Information about the number of independent experiments and quantitation is lacking. This affects the interpretation, as fun30 seems to affect the +3 nucleosome much more than let on in the description.

We appreciate the reviewer pointing out places in which our manuscript omitted key pieces of information (items 1 and 3), and we will fix these oversights in our revision.

With regard to point 2, we had written:

“Fun30 is also known to play a role in the DNA damage response; specifically, phosphorylation of Fun30 on S20 and S28 by CDK1 targets Fun30 to sites of DNA damage, where it promotes DNA resection (Chen et al. 2016; Bantele et al. 2017). To determine whether the replication phenotype that we observed might be a consequence of Fun30's role in the DNA damage response, we tested non-phosphorylatable mutants for the ability to suppress early replication of the rDNA in sir2; these mutations had no effect on the replication phenotype (Figure 2B), arguing against a primary role for Fun30

in DNA damage repair that somehow manifests itself in replication.”

We will expand on this to clarify our point in the revision.

<https://doi.org/10.7554/eLife.97438.1.sa4>